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# Spatial econometric analysis of the digital divide in Thailand at the sub-district level: Patterns and determinants☆

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ABSTRACT

This study explores spatial patterns of the digital divide and its determinants at the sub-district level in Thailand, uncovering nuanced patterns often overlooked in the in prior literature. To our knowledge, no studies in the digital divide field have offered such detailed estimations. Based on the Spatially Aware Technology Utilization Model, we constructed a conceptual model tailored to the digital divide's geospatial context in Thailand. Using a spatial econometric approach, we analyzed the 2021 data set on basic household necessities provided by Thailand's Community Development Department. The results show a positive spatial autocorrelation of the household internet access divide (HIAD) across Thailand's sub-districts. This spatial clustering tended to be high–high in the sub-districts located in the northeastern and northern regions, with low–low agglomerations in the central region of Thailand. In addition, the spatial econometric analysis results indicated that the HIAD may be influenced by various factors including demographic, economic, educational, government ICT prioritization, social capital, transportation, and disaster-related variables. This study provides both theoretical and policy implications to enrich geographic knowledge of the digital divide, specifically in the context of developing countries.

## 1. Introduction

During the COVID-19 pandemic, information and communication technologies (ICTs) were widely recognized for their capacity to rapidly and broadly distribute government information to the people. Their benefits also enabled many governments to effectively deliver public goods and services. In this way, ICTs have enhanced the ability to perform everyday online activities in several areas, including e-commerce, e-government, entertainment, healthcare, education, and social media. Despite the increased reliance on the Internet for societal engagement, a significant portion of the global population remains digitally disconnected. As of 2023, about 33% of people worldwide, predominantly in developing countries, do not have Internet access. This disparity is even more pronounced in the least developed countries, where only 36% of the population accessed the Internet, compared to a global average of 67% (ITU, 2023). This phenomenon is encapsulated in the concept of the "digital divide," which refers to the unequal access to the Internet across

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individuals, households, businesses, and geographical areas (OECD, 2001).

In Thailand, the scenario is gradually changing. Recent data from the National Statistical Office of Thailand (2021) indicates that approximately 83.8% of Thais now have Internet access. However, disparities persist, with regions like Bangkok boasting higher connectivity (93.8%) compared to other areas. Notably, the lack of access to the Internet and digital devices has hindered many from reaping the benefits of government initiatives and public services. This was evident during the Rao Chana ("We Win") scheme (a financial aid program for the poor implemented during the COVID-19 pandemic) and in the challenges faced by students in remote areas regarding online learning (Bangkok Post, 2021a; 2021b). For example, the Rao Chana was heavily criticized by many people who found it difficult to take advantage of the program's benefits due to the requirement that people register online. Digitally disconnected people living in rural and/or mountain areas requested the option to register on site at the nearest government bank branch, which incurs greater travel expenses and takes additional time (Wilson, 2021). Another example is online study. Students living in remote areas have described experiencing barriers to online learning during the pandemic, including device affordability issues as well as a lack of high-speed connectivity or any Internet signal at all (Bangkok Post, 2021b). Such examples highlight the multifaceted nature of the digital divide, which is further exacerbated in emerging markets by challenges including affordability, lack of infrastructure, and regulatory barriers. Collectively, these factors impede internet accessibility and socio-economic growth (Ashraf et al., 2015). Rural areas face particularly significant hurdles due to technical and monetary obstacles and neglect, limiting access to essential services such as healthcare, education, and government support (Elkhatib et al., 2015). This disparity is evident in Thailand and is manifested in the uneven distribution of IT hardware, internet connectivity, and digital skills between urban and rural regions (Nanthakorn et al., 2022). Moreover, internet penetration—or the lack thereof—significantly influences production outputs, especially in areas with lower internet usage rates (Punyasavatsut, 2019), underscoring the digital divide as a critical barrier to equitable development.

Recently, the Thai government has recognized the important role of digital technology in driving economic development and social equity. Policies such as the Digital Economy and Society Development Plan, the Digital Economy Promotion Master Plan, and the Thailand 4.0 initiative underscore the country's commitment to leveraging digital innovation for national advancement (Jongwanich, 2023, 2024, p. 80; OECD, 2022). These policies aim to transform Thailand into a digital technology-driven economy by integrating digital tools into all socio-economic activities to cultivate a value-based economy that prioritizes creativity, innovation, and the protection of intellectual property. Central to these initiatives is the goal of digital inclusivity, which ensures that all citizens can benefit from digital technologies, regardless of their socio-economic status or geographical location. In support of these goals, the Thai government launched the Village Broadband Internet (Net Pracharat) Project in 2017, aiming to extend high-speed internet access throughout rural Thailand. This is in alignment with the Thailand 4.0 policy's emphasis on innovation and reducing digital inequality. As of December 2017, the initiative has successfully installed fiber optic cable networks and provided free public Wi-Fi hotspots in 24, 700 rural villages. This expansion has garnered approximately 6.6 million registered users by July 2019. Training programs have educated over a million locals on internet use, enhancing their opportunities in education, healthcare, e-commerce, and government services. The project fosters sustainable development, propelling Thailand towards long-term stability, prosperity, and sustainability by equipping remote communities with crucial digital infrastructure (Asia-Pacific Telecommunity, 2019). However, the implementation of these digital divide policies faces considerable challenges. Infrastructure disparities, varying levels of digital literacy, and economic constraints hinder the equitable distribution of digital benefits. Despite being the backbone of Thailand's agricultural sector, rural areas often lack access to high-speed internet and digital services, limiting their integration into the digital economy. The situation is further complicated by Thailand's socio-economic landscape, which is characterized by income disparities and uneven development across regions (Barbier, 2023; Jongwanich, 2023; Nguyen et al., 2022).

These cases and the existing empirical evidence indicate that the digital divide is a complex and multilayered phenomenon. It is conceptualized as comprising three layers: the internet access divide (first level), the digital skills divide (second level), and the digital outcomes divide (third level; van Dijk, 2020). Geographical restrictions seem to be related to the widening of the digital divide at all levels. The size of a country and the presence of island and mountain areas may affect internet coverage. In addition, the level of technological development in a given area tends to resemble the level in neighboring areas, suggesting a spatial dimension to the digital divide. Such complexities highlight the need for a nuanced understanding of this issue, particularly concerning how it varies across different layers and geographic areas. Until recently, little has been known about the digital divide from a geospatial perspective.

Sarkar et al. (2015) attempted to examine the spatial clustering of ICT utilization across African countries and its driving factors. They revealed that Northern and Southern African countries have high ICT utilization, while most of sub-Saharan, Central, and West Africa have low ICT utilization. They also suggested that economic, demographic, and social openness are related to ICT utilization. In the same vein, Lin et al. (2017) revealed that income inequality was associated with the digital divide as measured in terms of Internet access, particularly in high-income countries. At the country level, Song et al. (2020) investigated the factors and spatial characteristics of the digital divide at the prefectural level in China. They revealed a positive spatial autocorrelation at all digital divide levels and found that residential income (in both urban and rural areas) and education levels were the leading determinants of China's prefecture-level digital divide. However, there are no studies to date analyzing the digital divide at a geographical unit smaller than the prefecture.

This study aims to complement this gap by examining the digital divide at the sub-district level in Thailand, offering a granular perspective that is often obscured in broader analyses. These detailed insights enable policymakers to pinpoint specific localities that require intervention. Whether the challenge is digital literacy or infrastructure, targeted solutions can be crafted to address these regional concerns. Policymakers particularly benefit from this precision because it makes it possible to allocate resources effectively to those areas that need it most. In addition, understanding influences between neighboring sub-districts can provide insights into how they might support or hinder each other's digital development. At this level, community-specific elements become more prominent, encouraging community engagement for improved adoption of digital inclusivity measures. A sub-district analysis makes sure that these important nuances are highlighted, paving the way for effective interventions in developing countries where disparities can be stark even within small regions. Therefore, exploring such a spatial distribution of the digital divide and its driving factors at the sub-district level would both enable us to explicitly understand the digital divide phenomenon from a geospatial perspective and provide more specific insights and policy implications for bridging the digital divide, especially in the context of developing countries like Thailand. Therefore, this study seeks to examine two main research questions.

(1) What are the spatial distribution characteristics of the digital divide at the sub-district level in Thailand?
(2) What are the factors that contribute to the digital divide at the sub-district level in Thailand?

Following this introduction, Section 2 lays the review of literature. Section 3 introduces a conceptual model of the digital divide. The methodology and data used are described in Section 4, while Section 5 presents the findings from our spatial econometric analysis. The paper concludes with a discussion of the implications of these findings for policy and practice.

## 2. Literature review

### 2.1. *Overview of the digital divide in developing countries*

The digital divide is a significant barrier to socio-economic development in developing countries and leads to notable disadvantages in the economic, political, and social realms. According to Chang et al. (2015), this divide critically impacts access to digital technologies, which in turn affects key societal aspects such as education, employment, and overall economic growth. The contributors to this divide are multifaceted, encompassing infrastructural, political, cultural, demographic, generational, and socio-economic factors (Ali & Faroque, 2023). The availability of high-speed broadband infrastructure is essential for attaining sustainable development goals, yet the gap in digital access is particularly pronounced in developing regions (Masonta, 2023). Geographic disparities, varying motivations for ICT usage, and differing rates of technology adoption further intensify this divide (Qadikolaei et al., 2024). Economic and socio-demographic factors are primary predictors of the digital divide, with the relative cost of internet access constituting a major hurdle in narrowing the disparity between lower and higher-income countries (Nielsen et al., 2018).

Recent statistics have revealed a significant disparity in internet penetration; while about 67% of the global population (5.4 billion people) were connected to the internet in 2023, only approximately 33% of individuals in developing countries had access to the internet (ITU, 2023). Despite this, connectivity in these nations has increased by around 17% over the past year, highlighting a significant but gradual improvement in access. Yet, even in areas covered by mobile broadband, a substantial "usage gap" persists, with nearly half the population in these regions remaining unconnected. This gap stems from various hurdles, including affordability, lack of digital skills, and limited availability of relevant content (ITU, 2023). In least-developed countries (LDCs), internet usage is alarmingly low at about 36%, with 17% of LDCs completely lacking mobile broadband coverage (ITU, 2022). Mobile technology is the predominant method of internet access in these regions, suggesting a focus for future development and support. Efforts to narrow the digital divide have had some success, yet affordable access continues to pose a challenge. In LDCs, the cost of a basic mobile broadband plan can constitute nearly 6% of the average income, which significantly exceeds the 2% affordability target set by the UN Broadband Commission (ITU, 2022). These statistics underscore the critical challenges of affordability and accessibility, with the combination of costly internet services and insufficient digital infrastructure severely limiting opportunities for economic growth, educational achievements, and socio-political participation in these regions.

### 2.2. *Internet access in developing countries*

Internet access in developing countries is constrained by a complex interplay of infrastructural challenges and geographical disparities that hinder equitable digital inclusion. These issues collectively underscore the pressing need for targeted policy interventions and infrastructural investments to bridge the digital divide.

#### 2.2.1. *Infrastructure challenges*

ICT infrastructure plays a critical role in shaping internet access within developing nations. Broadband connectivity, a recognized driver of socio-economic development, is predominantly concentrated in urban areas, thus exacerbating the digital divide between urban and rural communities. According to Ngwenya et al. (2023), the slow pace of internet penetration and digital service expansion in rural areas can be attributed to a combination of low average income, poverty, underdeveloped infrastructure, illiteracy, and unemployment. Moreover, the availability and reliability of the electricity supply are crucial for internet adoption; the absence of stable power is a significant barrier to internet access in less affluent regions (Armey & Hosman, 2016). Frischtak (2019) suggested that to counter these challenges, governments should foster an open and competitive market environment to attract the necessary infrastructure investments, thereby ensuring enhanced service quality and reduced costs. However, the development of ICT infrastructure often faces hurdles such as unreliable electricity supplies, which further complicates the accessibility of internet services (Armey & Hosman, 2016). Additionally, the spatial interdependence between countries highlights that internet access issues are interconnected across borders, requiring coordinated international policy efforts to address broadband subscription disparities effectively (Myovella et al., 2021).

#### 2.2.2. Geographical disparities

Geographical disparities, particularly the rural–urban divide, profoundly impact internet availability and access in developing countries. Rural areas frequently face limited connectivity opportunities due to inadequate wired communication infrastructure, including the frequent unavailability of fiber-to-the-home (FTTH) and very high bit-rate digital subscriber lines (VDSLs) in these regions (Thakur & Prasad, 2021). While cellular access technology has emerged as the primary broadband mechanism in these countries, its expansion in rural settings is hindered by economic challenges, including low average revenue per user and the difficulties associated with servicing sparsely populated areas (Thakur & Prasad, 2021). This lack of connectivity not only perpetuates the digital divide but also exacerbates disparities in internet access based on age, gender, income, and education levels, as highlighted by Sujarwoto and Tampubolon (2016). This uneven access underscores the necessity for comprehensive strategies that address the challenges in both infrastructure and service distribution and aim to enhance digital inclusivity across diverse regions.

### 2.3. Role of ICT policies in shaping internet accessibility

In developing countries, effective ICT policies are crucial for shaping internet accessibility and bridging the digital divide. The quality of governance significantly influences the efficacy of these policies, particularly in supporting inclusive, pro-poor development goals and enhancing ICT penetration across all layers of society. Barry (2018) highlighted that transparent and accountable governance is essential for the equitable distribution of technological benefits, especially for underprivileged populations. Investments in infrastructure are another critical component of ICT policies. Moreover, innovative and cost-effective interventions are necessary to close the digital divide, as noted by Masonta (2023). Such measures include rolling out affordable broadband services to rural and remote areas, developing sustainable models for community-owned broadband networks, and ensuring that policy frameworks support these initiatives.

However, ICT policies also face several challenges that impact their effectiveness. Qadikolaei et al. (2024) and Ohemeng and Ofosu-Adarkwa (2014) have discussed how disparities in access to and utilization of ICT exacerbate the digital divide, with factors like family income, the presence of research and development centers, and social exclusion significantly affecting this issue. Additionally, ICT, including access to mobile phones, internet use, and broadband connectivity, has been identified as a key driver of economic growth in developing countries (Bahrini & Qaffas, 2019; Farhadi & Ismail, 2014), emphasizing the need for increased investments in ICT infrastructure. Geographic disparities, varying motivations for ICT usage, and differential technology adoption rates within and across nations further complicate the effective implementation of ICT policies, especially for marginalized socioeconomic groups (Qadikolaei et al., 2024). Odongo (2014) pointed out that challenges such as inadequate infrastructure, high access costs, inappropriate policy regimes, and inefficiencies in network telecommunications provision remain substantial barriers to effective digital inclusion in developing countries.

This body of literature has highlighted a critical interplay between the availability, accessibility, and affordability of ICT and the relevant ICT policies in shaping internet access in developing countries. While it is evident that significant strides have been made in understanding the broader impacts of the digital divide, particularly in terms of socio-economic development, the existing research has lacked discussion on the current policies addressing these issues and the existing state of infrastructure.

### 2.4. Theoretical frameworks and models

The digital divide is a complex, multilayered phenomenon comprised of several indicators (variables). Pick and Sarkar (2016) summarized the four theories that have typically been used in past studies related to the digital divide. The first, the adoption-diffusion theory (ADT), was developed to understand how innovations are adopted by and diffused among potential users over time (Rogers, 1983). This theory is not limited to ICT but also involves innovations in healthcare, transportation, and management. In digital divide research, this theory has typically been applied to analyze ICT adoption and diffusion, and most work to date has focused on ICT accessibility rather than ICT usage. One weakness of this theory is that it does not capture the social and behavioral forces that affect individual adoption decisions (Pick & Sarkar, 2016). To address this shortcoming, several attempts have been made to complement ADT. For example, Asrani and Kar (2022) employed ADT to examine technological adoption in India and concluded that socioeconomic factors and urbanization are associated with computer adoption. They claim that these findings complement and support ADT. Likewise, Rice and Pearce (2015) integrated ADT into digital divide research by investigating mobile phone adoption at the individual level in the context of a developing country. The results of their model showed that economic conditions, household size, age, education, and internet access were significantly correlated with the digital divide.

The second relevant theory is the resources and appropriation theory (RAT), also well known as "van Dijk's theory" (van Dijk, 2020). It highlights inequalities in individuals' resources that may affect their ability to access the digital world and result in disparities with regard to their participation in society. This theory recognizes three levels of the digital divide, namely the internet access divide (the first level), the digital skills divide (the second level), and the digital outcomes divide (the third level). Pick and Sarkar (2016) noted that RAT can provide deeper theoretical insights than ADT because it incorporates more inclusive indicators, such as economic, social, cultural, and behavioral variables. However, the complexity of the theory means that the unit of analysis is intermixed in the results and collecting data is difficult. This theory is appropriate for case study research and was developed based on case studies and surveys conducted in developed countries. For instance, van Deursen et al. (2021) employed RAT as a research framework to investigate inequality regarding the accessibility of the Internet of Things (IoT) among the Dutch population and revealed that individual attitudes, education, economic conditions, and material access impacted IoT inequality. Similarly, van Deursen (2020) applied RAT to examine internet use and outcomes during the COVID-19 pandemic, and focusing on the determinants of digital skills among professional workers, van Laar et al. (2019) highlighted social support and training as potential factors for digital skills. Overall, RAT is more comprehensive than ADT and was specifically developed to investigate the digital divide phenomenon.

The unified theory of acceptance and use of technology (UTAUT) is the third useful theory for explaining the digital divide, albeit in terms of technology acceptance and usage rather than access. This theory was originally developed by Venkatesh et al. (2003) and can be used in digital divide research to explain individuals' behavioral intentions toward using ICT, treating these intentions as the dependent variable. The four main constructs or moderators for the theory are attitude, effort expectancy, performance expectancy, and social influence. As this theory is mainly focused on the individual level, it is rarely applied to larger units of analysis. Example applications include the case of the belief in internet usage among Americans; Porter and Donthu (2006) found that age, economic conditions, education, and race were correlated with differences in internet use beliefs. Niehaves and Plattfaut (2014) implemented this framework to examine the use of the internet among older people in Germany. Likewise, Sipior et al. (2011) explored the factors correlated with the use of e-government services among American citizens and likened disparities in usage to the digital divide phenomenon. These studies deomonstrate the varied applications of UTAUT in digital divide research, especially at the individual level.

The fourth model mentioned by Pick and Sarkar (2016) was their own. They argued for the importance of their model, which integrates a geospatial dimension into digital divide research. They termed it the Spatially Aware Technology Utilization Model (SATUM) and first proposed it as a theoretical model in their book *The Global Digital Divides: Explaining Change* (Pick & Sarkar, 2015). SATUM employs a set of technology utilization factors as dependent variables (e.g., broadband internet subscribers, fixed phone subscribers, internet users, mobile telephone subscribers, personal computers, secure internet servers, and social media) and a set of social, economic, governmental, and innovation factors as independent variables (e.g., demographics, ethnicity, economic status, education, government prioritization of ICT, infrastructure, innovation, social capital, and societal openness). It should be noted that SATUM does not consider psychological aspects. The relationships between the dependent and independent variables should be screened for the existence of spatial dependence. Depending on data availability, this model can be applied over various units of analysis, including the global, national, state, provincial, and prefectural levels, as well as even smaller levels like districts, sub-districts, and villages. However, there has been limited research attempting to understand the digital divide phenomenon from a geospatial perspective. Researchers have explored the case of internet broadband access in the United States in terms of several aspects, including spatial inequality in broadband access (Grubesic & Murray, 2002), spatial distribution of internet broadband providers (Grubesic, 2008), availability of advanced telecommunication services (Grubesic & Murray, 2004), and predictions for broadband provision generally (Mack & Grubesic, 2009) and in tribal areas (Mack et al., 2022). SATUM has also been applied to investigate the spatial patterns in the digital divide at several levels. For example, Pick and Nishida (2015) investigated technology utilization at a global scale. At a regional level, Pick et al., (2021) conducted a spatial analysis of the digital divide in Latin America and the Caribbean, and Myovella et al. (2021) studied the digital divide across sub-Saharan African countries. At the state or prefecture-level, researchers have investigated the digital divide at the prefecture level in Japan (Nishida et al., 2014) and China (Song et al., 2020) and at the state level in the United States (Pick et al., 2015).

Despite the comprehensive exploration of the digital divide in developing countries that has been presented in the literature, several critical gaps remain, particularly concerning the level of geographic granularity and the specificity of analyses. Most existing studies have predominantly focused on the national or regional levels, significantly overlooking the intricate variations that emerge at the sub-district level. This oversight is crucial as it results in failure to capture the local impacts of ICT policies and infrastructure initiatives, which can vary dramatically within a country. Furthermore, while geographical disparities have been acknowledged in previous research, there is a noticeable lack of detailed exploration at the sub-district level, where variations in internet accessibility and usage can be particularly pronounced. Additionally, spatial econometric analysis has not been extensively applied to examine the digital divide. This technique are essential for understanding the correlation between digital divide factors and spatial elements and identifying specific areas where interventions are most urgently needed. Moreover, the existing literature often lacks sufficient empirical evidence from developing countries like Thailand. In summary, although existing literature and theories have significantly contributed to understanding the digital divide, there remains a notable gap in the research resulting from the lack of studies exploring this phenomenon at more granular geographic scales, especially in developing countries. To date, no research has examined the digital divide at geographical units smaller than the prefecture level. This study aims to address this gap by employing the SATUM to analyze the digital divide at the sub-district level in Thailand. By doing so, it provides a unique perspective on the spatial aspects of the digital divide, aiming to enhance targeted policy interventions and infrastructure planning at a more localized level.

## 3. Conceptual model of the digital divide

This study's design is based on the conceptual model of the digital divide from a geospatial perspective that has been proposed in previous studies (Myovella et al., 2021; Nishida et al., 2014; Pick et al., 2015; Pick & Sarkar, 2015, 2021; Song et al., 2020). We modified the SATUM to analyze the digital divide between geographical areas (in this study, sub-district areas in Thailand). This model enabled us to estimate the spatial correlation between posited variables while taking into consideration data availability at the sub-district level in Thailand. In the administrative hierarchy of Thailand, a sub-district, known as a "Tambon (ตำบล)" in Thai, is divided into multiple villages or "Muban (หมู่บ้าน)." Each Muban represents the smallest administrative subdivision in Thailand. While a Tambon may handle broader administrative and governance functions for its region, individual Muban focus on local community-based affairs specific to their village.

In our conceptual model, the Household Internet Access Divide (HIAD) is positioned as the dependent variable, while the independent variables are organized into two primary groups: factors related to the Availability of ICT Services and the Capacity of ICT

Usages. These variables are subdivided into seven categories consistent with the framework proposed in SATUM: demographic, economic, educational, government ICT prioritization, social capital, transportation, and disaster-related factors. Our conceptual model, illustrated in Fig. 1, depicts the expected relationships between these variables based on the literature. It is important to note that due to data constraints, we were unable to include the categories of infrastructure, innovation, and societal openness as originally outlined in SATUM. Additionally, the incorporation of transportation and disaster-related variables expands the SATUM framework, introducing new areas that are expected to significantly enhance the understanding and analysis of the digital divide. The subsequent subsections justify the selection of dependent and independent variables and align them with the established categories within the SATUM framework. We also explain the causal pathways that link all independent variables to the HIAD, categorizing them under the overarching sets of Availability of ICT Services and Capacity of ICT Usages, thereby clarifying their roles and interconnections within our model.

### *3.1. Household internet access divide (HIAD)as dependent variable*

As a proxy for the digital divide phenomenon at the sub-district level, we used the HIAD as the main dependent variable. The digital divide in this study is the difference in Internet access at the household level across sub-districts, which could lead to socioeconomic development imbalance. Accordingly, the HIAD variable is calculated as the ratio of households without an Internet connection to the total number of households. We expected the existence of spatial agglomeration of the digital divide across sub-districts in Thailand, which may be determined by several potential factors. It should be noted that in Thailand, utilizing the HIAD as the proxy of the digital divide is beneficial and limiting. HIAD offers a straightforward and quantifiable measure, capturing the digital divide at the household level in Thailand. Household internet access has direct implications for economic activities such as e-governance, e-commerce, remote work, and online education. Using HIAD can thus provide insights into potential economic disparities.

However, HIAD might oversimplify the digital divide as it represents a multi-layered phenomenon comprising three levels (i.e., digital access, skills, and outcomes). While HIAD captures Internet access, it doesn't reflect the quality, speed, or reliability of connections. It also fails to encompass digital skills and the actual benefits derived from the digital usage, potentially representing passive access without effective use. Furthermore, HIAD provides a static snapshot and does not reflect how internet access is changing as technology develops. Importantly, it is crucial to recognize that the HIAD metric cannot fully distinguish between a lack of internet access due to infrastructure unavailability versus individual or household choices not to adopt the technology, even if accessible. This distinction is a limitation of our study. Even though HIAD has some limitations in terms of accurately capturing the complex nuances of the digital divide, we believe that its advantages and capacity to offer a concrete, quantifiable measure make it an important metric for our study. We have chosen to use HIAD as the main measure for comprehending the digital divide in the context of Thailand due to the data available and the need for a simple, understandable metric that can be easily related to policymakers and the general public.

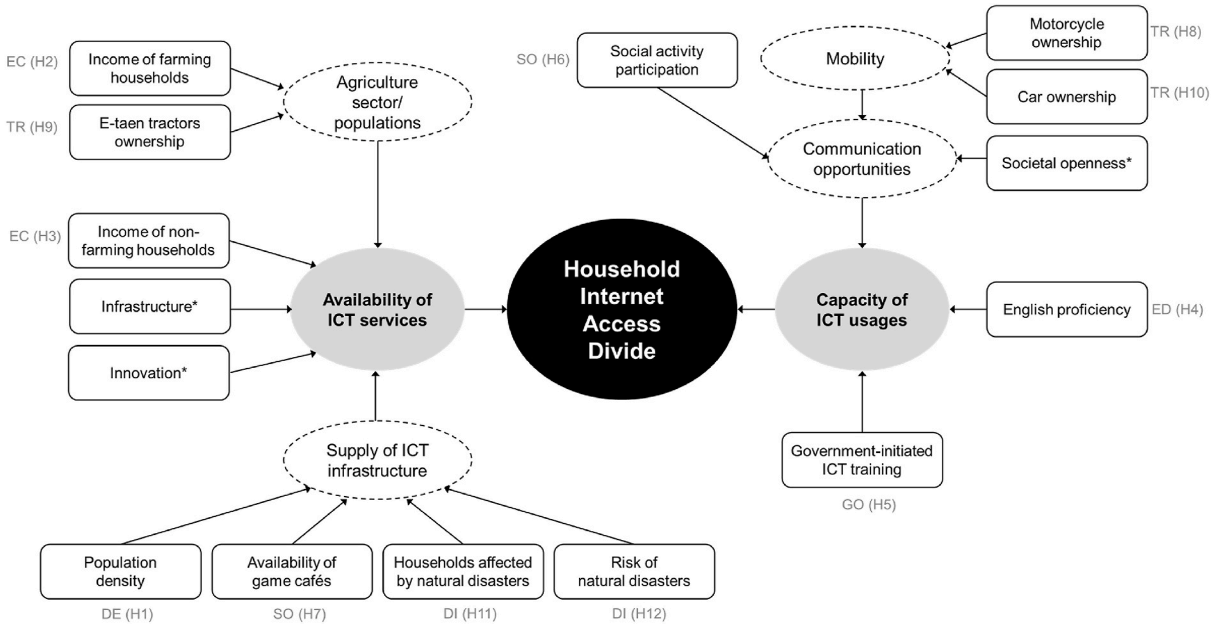



**Fig. 1.** Conceptual model of the digital divide in Thailand based on SATUM. Note: Abbreviations used for the categories within the model are Demographic (DE), Economic (EC), Education (ED), Government Prioritization of ICT (GO), Social Capital (SO), Transportation (TR), and Disasters (DI). Each category is linked to specific hypotheses, indicated by the hypothesis numbers in parentheses following each category's abbreviation. An asterisk (*) represent correlations that we were unable to estimate in this study due to the limited availability of data on Thailand.

### 3.2. Independent variables

#### 3.2.1. Demographic

Demographic characteristics significantly shape the landscape of internet access and contribute to the digital divide. In developing countries, the diversity of these factors presents unique challenges and opportunities for digital inclusion. Research by Nishida et al. (2014), Pick et al. (2015), and Song et al. (2020) have suggested that areas with higher population density often exhibit characteristics conducive to better internet access. These include greater economic activity, which can encourage investment in ICT infrastructure, and denser social networks, which might increase demand for communication technologies. Additionally, the proximity of potential customers in high-density areas can reduce service provision costs. Therefore, while these past studies used demographics such as urban population, race, and working-age population, this study focuses on population density as it is particularly pertinent in the context of Thailand. The country's mix of densely populated urban centers and sparser rural regions likely leads to uneven internet access across different regions. Government policies and initiatives concerning ICT development might be affected by population density, favoring regions perceived as more economically attractive due to their greater population concentrations. Furthermore, recent insights by Zhou et al. (2022) highlighted the dynamic interplay between population characteristics and the digital economy. They noted that regions with robust digital infrastructure can attract residents, whereas those lacking digital amenities might experience population declines.

Accordingly, **Hypothesis 1 (H1)** in this study posits a negative relationship between population density and HIAD within Thai sub-districts. Sub-districts with higher population density are hypothesized to experience a narrower HIAD. This relationship will be quantified by measuring population density as the natural logarithm of the number of individuals per square kilometer.

#### 3.2.2. Economic

Economic factors play a pivotal role in digital divide research. Household income, per capita income, and employment rates are traditional indicators of economic influence on digital adoption. Past studies have emphasized the critical link between economic status and internet access, highlighting how greater economic resources enable enhanced digital access through increased connectivity and device affordability and present the potential for economic returns from internet usage (Pick et al., 2021; Setthasuravich & Kato, 2020, 2022; Song et al., 2020; Szeles, 2018). Further exploration of economic factors reveals that internet access in developing regions like Thailand is influenced by multiple dimensions. The affordability of internet services, emphasized by researchers such as Ashraf et al. (2015) and Vincent (2016), remains a pivotal factor. Additionally, Ordóñez et al. (2016) argued that the reciprocal relationship between economic development and internet adoption suggests that increased internet usage can stimulate economic activities and vice versa. The interconnection of these factors underscores the importance of broadband infrastructure and regulatory environments conducive to ICT investment, as highlighted by Bánhidi (2021) and Dkhil and Jebsi (2020).

The specific economic conditions in Thailand further shape this inquiry. The Thai economy is characterized by a pronounced disparity between urban and rural areas, particularly between farming and non-farming households. Farming households, primarily based in rural areas, may have not only different income levels but also distinct perceptions of the utility of internet services and digital infrastructure at various levels (Wims, 2014). This nuanced context informs two distinct hypotheses. **Hypothesis 2 (H2)** posits a negative relationship between the income of farming households (measured as the natural logarithm of average annual income per farming household) and HIAD in Thai sub-districts. Additionally, **Hypothesis 3 (H3)** proposes a positive relationship between the income of non-farming households (measured as the natural logarithm of average annual income per non-farming household) and HIAD in Thai sub-districts.

#### 3.2.3. Education

Education-related factors have often been shown to correlate with the digital divide. Several studies have utilized variables such as household expenditure on education (Nishida et al., 2014), use of English as a primary language (Pick et al., 2021), tertiary educational attainment (Szeles, 2018), and adult literacy rates (Song et al., 2020). Previous literature has emphasized the importance of education systems and technological training in equipping populations with the necessary skills for internet access. Song et al. (2020) highlighted the role of secondary education in improving ICT outcomes, suggesting that increased educational investment can enhance technology adoption. Similarly, Billon et al. (2021) noted that educational disparities can negatively impact internet use, particularly in countries with middle-income economies. These findings underscore the need for policies that not only expand physical access to technology but also improve educational outcomes related to digital skills to foster a more inclusive digital environment.

In general, English is the language used in digital technologies and media. Although basic information and functions (e.g., email) are widely available in local languages, English comprehension capability is needed to access the broader digital world and related activities (Murphy & Arciuli, 2024). Proficiency in English is posited to enhance the ability to use digital devices, access information, and leverage the benefits of the internet effectively (Alsmari, 2021). In Thailand, English is not the primary language, but it is used extensively in global technology, commerce, and online content. As a developing nation, Thailand faces notable educational disparities between urban and rural areas, which are likely mirrored in the levels of English proficiency and, consequently, internet access (Khan et al., 2019; Lee, 2022). Regional differences in government initiatives and educational policies could also influence the priority given to English language learning, reflecting the economic potentials of different areas (Kirkpatrick & Barnawi, 2017). The selection of English comprehension as an educational indicator is grounded in its role as a key tool in digital technologies and media and reflects an understanding that English proficiency is a critical educational factor that could influence the digital divide in Thailand. Therefore, **Hypothesis 4 (H4)** posits a negative relationship between English proficiency (measured as the proportion of the total population made up by individuals aged 15–59 who can read and write in English) and HIAD in Thai sub-districts.

#### 3.2.4. Government prioritization of ICT

Government prioritization of ICT or support from the government regarding ICT use has been identified as an important factor in narrowing the digital divide. Such measures encourage people to learn the basics of ICT use and its applications (Pick & Azari, 2011). These policies are particularly critical in proactive efforts to enhance digital literacy and accessibility, and they play a pivotal role in mitigating the barriers to digital adoption. Government-led ICT training programs, especially those targeting underserved communities, are crucial for reducing the digital divide (Chohan & Hu, 2022). The Thai government has actively implemented programs designed to improve digital skills across the population (Jantavongso, 2022). However, it is expected that the distribution and emphasis of these initiatives varies significantly across different areas, often influenced by perceived economic potentials. This variation could lead to disparities in digital skills and internet access between areas that receive more focused governmental support and those that do not. Accordingly, **Hypothesis 5 (H5)** proposes a negative relationship between the degree of participation in government-initiated ICT training (measured as the natural logarithm of the total number of individuals participating in such programs) and HIAD in Thai sub-districts.

#### 3.2.5. Social capital

According to Pick and Sarkar (2015), social capital refers to the connections and interactions among individuals via the social and physical networks or organizations that support those individuals' interactions. Recently, Wang et al. (2021) found that social capital influences the digital divide; following social gatherings, individuals who have participated in social interactions often seek additional connections and further communication via digital platforms. In exploring the impact of social capital on the digital divide, particularly within the context of Thailand, we focus on two distinct but interconnected dimensions: participation in social activities and the presence of game cafés. These elements play a significant role in shaping the local digital landscape by facilitating digital skill development and community engagement.

Social activities are shared events that typically foster interactions among participants. These interactions can lead to the sharing of knowledge and can include conversations about digital skills and resources (Karna & Ko, 2015; Yuan, 2021). Furthermore, engaging in social activities creates a supportive environment where individuals feel more comfortable exploring and adopting new technologies with encouragement from their peers and community members (Ludwig et al., 2019; Pederzoli, 2019, pp. 395–408). When individuals interact within these settings, they are likely to discuss digital devices and online services and demonstrate their use. Active communities are better positioned to advocate for and support the expansion of digital infrastructure, which in turn facilitates improved internet access. Therefore, **Hypothesis 6 (H6)** posits that there is a negative relationship between social activity participation (measured as the proportion of households that participate in social activities) and HIAD in Thai sub-districts.

Given their role as social hubs where individuals, particularly youth, engage with digital technologies, game cafés provide more than just gaming opportunities; they offer informal tech support and serve as crucial community centers for digital interaction, particularly in areas with limited home internet access (Wongyai & Huttasin, 2022). Therefore, while game cafés foster digital engagement and learning through peer interactions, their presence may also reflect underlying deficiencies in home-based digital access. Consequently, **Hypothesis 7 (H7)** posits a positive relationship between the density of game cafés (measured as the natural logarithm of the total number of game cafés) and HIAD in Thai sub-districts.

#### 3.2.6. Transportation

The role of transportation in the digital divide is multifaceted. The ownership of specific types of vehicles can act as an indicator of economic well-being, reflecting varying levels of economic status. Beyond that, reliable transportation provides essential access to physical locations with internet resources, such as public ICT centers and internet cafés, thereby facilitating direct engagement with digital opportunities and services, which is especially critical in areas lacking robust internet infrastructure. Recent studies have indicated a broader connection between transportation and the digital divide, especially from the geospatial perspective (Setthasuravich & Kato, 2022). Additionally, Vural et al. (2020) suggested that the use of digital technology in transportation can mitigate barriers to access, but the process of digitalization faces hindrances due to the conservative nature of the transportation industry. Durand et al. (2022) further highlighted how difficulties in the digitalization of transportation services vary across demographic lines, underscoring disparities in access that can parallel those observed in digital connectivity. In Thailand, transportation methods vary significantly. *E*-taen tractors are prevalent in agricultural settings, whereas car ownership is more common in urban areas and may indicate higher socioeconomic status. Furthermore, motorcycles are widely used throughout the country. Each of these transportation types offers different implications for internet access. Ownership of cars, motorcycles, or farm tractors not only reflects an individual's mobility but also their access to resources (Bhui & Pandit, 2021; Fronteli & Pacheco Paladini, 2023; Galich & Stark, 2021), which can be crucial for digital accessibility.

Motorcycles are a widespread mode of transportation in Thailand, especially in rural areas (Peraphan et al., 2017). However, their direct effects in facilitating internet access are less evident. It is expected that households may invest in motorcycles and allocate a significant portion of their limited financial resources to transportation, potentially at the expense of other non-essential services like household internet connectivity. This trade-off is especially likely in less affluent areas where the necessity of transportation for work or education overshadows the perceived need for internet access (Forman et al., 2021, pp. 31–45). Moreover, Chiu (2023) found that motorcycle ownership is often more necessary and common in rural areas where public transportation is limited or non-existent. These areas might also lack the infrastructure required for stable and affordable internet access. Given this context, **Hypothesis 8 (H8)** posits that motorcycle ownership (measured as the natural logarithm of the total number of motorcycles) in Thai sub-districts has a significant effect on the HIAD.

*E*-taen tractors are fundamental to agricultural operations in Thailand and are predominantly used in rural areas where agriculture drives the economy (Hossen et al., 2020). In such regions, households often prioritize investment in essential agricultural equipment like *E*-taen tractors over digital technologies (Barbier, 2023). It is expected that the significant costs involved in purchasing and maintaining these tractors deplete financial resources that might otherwise be allocated to technological advancements, including internet services. In these contexts, the application of technology is generally directed towards agricultural enhancements rather than expanding digital connectivity. Given their specialized use, the direct relationship between *E*-taen tractor ownership and internet access is expected to be minimal, but it may nonetheless be reflected in the HIAD in rural areas that prioritize agriculture. It is therefore hypothesized that there is a positive relationship between *E*-taen tractor ownership (measured as the natural logarithm of total number of *E*-taen Thai farm tractors) and HIAD in Thai sub-districts, as posited in **Hypothesis 9 (H9)**.

Car ownership is typically higher in urban areas and correlates with higher socioeconomic status (Sun et al., 2024; Zhao & Bai, 2019). Additionally, cars enhance mobility, allowing individuals greater access to internet-enabled services and facilities that might not be available in the immediate vicinity (Hong et al., 2019). This access is crucial in semi-urban and underserved rural areas where fixed broadband services may be lacking. Accordingly, **Hypothesis 10 (H10)** posits a positive relationship between car ownership (measured as the natural logarithm of the total number of cars) and HIAD in Thai sub-districts.

### 3.2.7. *Disaster*

We added disaster variables to the model since the relationship between the digital divide and disasters is receiving increasing attention among researchers. Natural disasters can cause severe disruptions to communication networks, directly hindering internet access and exacerbating the digital divide (Dargin et al., 2021). Moreover, the recovery efforts following disasters often lead to a significant diversion of resources from digital infrastructure development to immediate relief activities, particularly impacting vulnerable areas (Madianou, 2015). However, the recurring nature of such events can heighten awareness about the need for resilient communication systems, potentially spurring investments aimed at enhancing digital infrastructure to improve disaster preparedness. Thailand's diverse geographic and climatic conditions expose it to a variety of natural disasters, including floods, droughts, and storms (Fakhruddin, 2024; Visessri & Ekkawatpanit, 2020; Phoompanich, Barr, & Gaulton, 2019). The impact of these disasters on the digital divide is significantly shaped by regional disparities in existing digital infrastructure and the effectiveness of government responses to these disasters (Marshall et al., 2023). We anticipated that areas affected by disasters or at greater risk of experiencing natural disasters find it more difficult to access the internet. This hypothesis is grounded in the observation that natural disasters can disrupt infrastructure, including communication networks, and thereby impede internet access. Moreover, the aftermath of such disasters often shifts the focus and resources of communities and governments away from digital infrastructure development towards immediate relief and recovery efforts. Therefore, we propose two further hypotheses. **Hypothesis 11** (H11) posits a positive relationship between the number of households affected by natural disasters (measured as the total number of households affected by natural disasters in the previous year) and the HIAD in Thai sub-districts. **Hypothesis 12** (H12) proposes that there is a positive relationship between the risk of natural disasters (measured as whether a sub-district is at risk of natural disasters) and HIAD.

It is important to acknowledge the potential for multicollinearity among our independent variables. Specifically, the indicators of economic, social capital, and transportation often exhibit intercorrelations. This potential issue will be carefully examined in subsequent stages of the analysis to ensure the robustness of our findings.

## 4. Methods and data

This study employed a spatial econometric approach. We expected to find a positive spatial autocorrelation of the internet access divide across Thailand's sub-districts, where similar values of the HIAD are geographically clustered in spatial proximity. First, we performed spatial autocorrelation tests to prove the presence of the spatial clustering of the HIAD. Following this, we used spatial regression models to determine the factors that contribute to the digital divide at the sub-district level in Thailand. To facilitate the analysis, a dataset of basic household necessities information in 2021 was provided by the Community Development Department of Thailand.

### 4.1. *Spatial weight matrix*

The spatial weight matrix is a tool for quantifying spatial relationships within a dataset, as noted by Anselin (2002). It employs two primary methods for construction: contiguity weights and distance weights. Distance weights articulate spatial relationships based on proximity, typically calculated using distances between geographic feature centroids. In contrast, contiguity weights are based on shared boundaries, leading to a binary matrix that delineates shared borders between areas. This matrix is instrumental in defining spatial connectivity within the study's framework. It can be expressed as follows:

$w_{ij} = 1$ if sub-district *i* shares its border with sub-district *j*.

0 otherwise (1)

In the contiguity matrix $w_{ij}$ the value is set to 1 if two areas share a boundary (are contiguous), and 0 otherwise. The interpretation of "neighbor" affects the outcome, with rook contiguity and queen contiguity being the two primary methods for determining neighboring relationships in sub-district polygons. Rook contiguity is based on shared edges, while queen contiguity includes both shared edges and vertices.

The queen contiguity criterion considers a common edge or common vertex between features, while the rook contiguity criterion only defines neighbors as two spatial features that share a shared edge. To create a neighbors list, this study used queen contiguity weights as the most adequate spatial weighting method, since the study focuses on a location pattern of polygon feature classes with organic forms and various sizes. Row standardization was used to eliminate bias related to the distribution of neighborhood size. Each weight is divided by its row sum to make sure every row of the matrix sums up to 1.

### 4.2. Spatial autocorrelation tests

We employed Exploratory Spatial Data Analysis (ESDA) to examine if the digital divide in Internet accessibility within Thailand is geographically clustered or dispersed, as suggested by Haining et al. (1998). To achieve this, we calculated the global Moran's $I$, a measure that quantifies the degree of spatial autocorrelation present in the data. The global Moran's $I$ index, ranging from −1 to 1, measures spatial autocorrelation. A value near 1 signifies positive autocorrelation, while a negative value indicates negative autocorrelation. A value close to 0 suggests a random distribution with no spatial correlation between sub-districts. Although this index reveals the presence of positive or negative spatial autocorrelations in the outcomes for each sub-district and its neighbors, it does not specify the locations of these clusters. It can be calculated as follows:

$$I = \frac{n\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(x_i - \overline{x})(x_j - \overline{x})}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}\sum_{i=1}^{n}(x_i - \overline{x})^2} \tag{2}$$

where $I$ denotes the global Moran's $I$ statistic, $n$ is the total number of sub-districts, $x_i$ and $x_j$ denote the HIAD value of sub-districts $i$ and $j$, $\overline{x}$ is the average value of the HIAD for all sub-districts, and $w_{ij}$ represents the spatial weight matrix.

To further examine the observed variable in each location, we applied local spatial autocorrelation analysis. This detailed approach assesses whether the characteristics of a location are similar to or differ from those in its surrounding areas. For measuring local spatial autocorrelation, we employed tools such as the Moran scatterplot and local indicators of spatial association (LISA), as outlined by Anselin et al. (2010). These methods are instrumental in identifying and understanding localized spatial patterns. The local Moran's $I$ is a LISA statistic that is quite similar to the global Moran's $I$ index. The parameters defined in Equation (3) are used similarly in Equation (4). The local Moran's $I$ is expressed as follows:

$$I_i = \frac{(x_i - \overline{x})}{S_i^2}\sum_{j=1, j\neq i}^{n} w_{ij}(x_j - \overline{x}), \tag{3}$$

$$S_i^2 = \frac{\sum_{j=1, j\neq i}^{n}(x_j - \overline{x})^2}{n-1} \tag{4}$$

where $S_i^2$ is the variance of $x$ in sub-district $i$, $I_i$ denotes the local Moran's $I$ coefficient in sub-district $i$.

### 4.3. Spatial regression models

In order to analyze the determinants of the digital divide and their relationship with socioeconomic factors, spatial regression models are conventionally estimated using a maximum likelihood approach. We conducted a spatial lag model (SLM) and spatial error model (SEM).

The SLM is expressed as follows:

$$HIAD_{it} = \alpha_0 + \rho\sum\nolimits_{j=1}^{n} w_{ij}HIAD_{jt} + \beta_n IN_{it} + \varepsilon_{it} \tag{5}$$

where $HIAD_{it}$ is the HIAD, $\alpha_0$ is the intercept, $\rho$ is the spatial correlation coefficient, $\sum w_{ij}HIAD_{jt}$ is the spatially lagged variable, $\beta$ is the vector of coefficients, $IN_{it}$ is the vector of the independent variables, $\varepsilon_{it}$ is the error term, and $i$ and $t$ denote the sub-district and year.

The SEM is calculated as follows:

$$HIAD_{it} = \gamma_0 + \delta_n IN_{it} + \mu_{it}$$

$$\mu_{it} = \lambda\sum_{j=1}^{n} w_{ij}\mu_{it} + \phi_{it} \tag{6}$$

where $\lambda$ represents the spatially correlated error coefficient, $\delta$ is the vector of coefficients, $\gamma_0$ is the intercept, and $\phi_{it}$ is the error term.

### 4.4. Data and geoprocessing

To facilitate the analysis, a dataset of basic household necessities information in 2021 was provided by the Community

Development Department (CDD) of Thailand. The data were collected biennially at the village level throughout the country (with the exception of Bangkok). The original tabular datasets were then aggregated into a polygon of the administrative sub-district boundary shapefile through a geocoding process using the ArcGIS Pro software. In this study, the dependent variable is HIAD. A variety of independent variables were compiled.

Table 1 presents a comprehensive breakdown of potential variables leveraged in this study, along with their definitions and respective classifications in terms of demand or supply sides. The digital divide category is denoted by the dependent variable, "Household Internet access divide" (HIAD), which represents the proportion of households without internet connectivity relative to the total households. This metric seeks to quantify the levels of the digital divide in our study context.

Factors classified under the Demand side encompass variables that inherently reflect the population's needs or requirements for digital services: *Demographic*, we incorporate "Population density," which denotes the natural logarithm of the number of inhabitants per square kilometer. *Economic*, income measures, differentiated for farming and non-farming households, gauge the financial capacities of households in their ability to access digital services. *Education*, "English proficiency" serves as a proxy for digital accessibility, as a substantial part of the digital realm operates in English. *Social Capital*, variables related to "Social activity participation" and the presence of "Game cafés" offer insights into communal engagement with digital platforms. *Transportation*, this captures the presence of transportation assets like motorcycles, cars, and *E*-taen Thai farm tractors, potentially indicative of economic status and thus, the capacity for digital engagement. *Disaster*, metrics related to households affected by natural disasters and the inherent risk of natural disasters in sub-districts can hint at infrastructural and connectivity challenges.

On the supply side, *Government prioritization of ICT*, the variable, "Government-initiated ICT Training," is employed to quantify the proactive efforts of the government in bridging the digital divide by offering ICT training, thus representing the supply of digital skills. The incorporation of these variables, based on their respective classifications, seeks to offer a holistic understanding of the multifaceted dimensions of the digital divide in our context.

Most variables were considered continuous, [1]though the risk of natural disaster variable (*DisRisk*) is a dummy variable with a binary value of 0 or 1. Continuous variables in this study include both ratio and interval scales. [2]Ratio-scale variables were calculated as the ratio of the total population, with a minimum value of 0 and maximum value of 1. The interval-scale variables for socioeconomic-related factors, [3]as shown in Table 2, were heavily skewed with outliers, which should also be considered. To overcome this issue, a natural logarithm transformation method was adopted to convert interval-scale continuous variables closer to a normal distribution before performing a spatial regression model.

Table 2 presents the descriptive statistics of several variables integral to our study. The Household Internet Access Divide (HIAD), with a mean of 0.491 and median of 0.512, slightly leans left in distribution. Population Density (PopDens_L) averages at 4.320 and possesses a mild rightward skew with a skewness of 1.040. As for the income, the Income per Farming Household (IncFarm_L) displays a mean of 10.760, but its left-skewed nature is evident with a skewness of −3.430. Income for Non-farming Households (IncNFarm_L) has a comparable left-skew, as seen in its mean of 9.860 and skewness of −2.180. The capability in English, represented as English Proficiency (EngPro), yields an average score of 0.270. Government-initiated ICT Training (Gov_ICT_L) showcases a median value greater than its mean, indicating that the government's efforts in ICT training are widespread but may be more concentrated in certain areas. Social Activity Participation (SocActv) suggests a highly socially active populace, with an average participation of 76%. Interestingly, the Game Cafés (SocGame_L) metric shows a mean of 0.690 but a median of zero, hinting at the presence of regions abundant in game cafés while others completely lack them. Transportation metrics emphasize the importance of motorcycles, *E*-taen Thai farm tractors, and cars in Thai society, with the mean values of 7.160, 3.360, and 6.570 respectively. The data concerning natural disasters reveals two key insights: approximately 60% of households, as denoted by the Households affected by Natural Disasters (DisHH) metric, experienced repercussions from calamities, while the Risk of Natural Disasters (DisRisk) showcases that areas truly at risk are rare but significantly vulnerable, evident in the high skewness of 9.210. Lastly, we checked for multicollinearity among all explanatory variables by testing variance inflation factors (VIFs). The results showed that the VIF values for all explanatory variables ranged from 1.03 to 2.96 (i.e., less than 10), indicating that there are no issues of multicollinearity (Lin, 2008).

## 5. Results

### 5.1. Spatial patterns of household internet access divide (HIAD)

The analysis of HIAD across Thailand's sub-districts reveals significant spatial clustering patterns that are critical to understanding the digital divide in the country. By employing spatial autocorrelation tests on data collected at the village level. We then calculated

[1] The dependent variable (Digital divide) does not have a classification as it is the outcome being analyzed.

[2] A game café, also known as a "Net Café" or "Internet Café," is not only a place to access the internet in Thailand but also a well-liked gathering place where people congregate to play online games, use social media and engage in other digital activities, particularly the younger generation. These cafés provide computers equipped with the latest hardware and software, ensuring an optimal gaming experience (Wongyai & Huttasin, 2022). These gaming cafes frequently act as the main entry points into the digital world for many Thais due to the digital divide and the lack of personal computers in some households.

[3] *E*-taen is a common type of agricultural vehicle in Thailand. It is designed specifically for the varied terrain of Thai farmlands and resembles a tractor or a modified truck. Primarily used for farming activities, its design facilitates tasks like plowing and transportation of crops. Given its ubiquity in rural Thailand, it serves as a symbol of Thai agriculture (National Science and Technology Development Agency, 2011).

**Table 1**
Potential variables and definitions.

| Category | Type | Variable name | Abbreviation | Definition | Side | Source |
|---|---|---|---|---|---|---|
| Digital divide | Dependent | Household internet access divide | HIAD | Ratio of households without internet connection to total number of households | n/a | CDD |
| Demographic | Independent | Population density | PopDens_L | Natural logarithm of population density per square kilometer | Demand | CDD |
| Economic | Independent | Income per farming household | IncFarm_L | Natural logarithm of average annual income per farming household | Demand | CDD |
| | Independent | Income per non-farming household | IncNFarm_L | Natural logarithm of average annual income per non-farming household | Demand | CDD |
| Education | Independent | English proficiency | EngPro | Ratio of individuals aged 15–59 years able to read and write in English to total population | Demand | CDD |
| Government prioritization of ICT | Independent | Government-initiated ICT Training | Gov_ICT_L | Natural logarithm of total number of individuals who have participated in government-initiated ICT training | Supply | CDD |
| Social capital | Independent | Social activity participation | SocActv | Ratio of households in participating in social activities | Demand | CDD |
| | Independent | Game cafés | SocGame_L | Natural logarithm of total number of game cafés | Demand | CDD |
| Transportation | Independent | Motorcycles | Tran_MTB_L | Natural logarithm of total number of motorcycles | Demand | CDD |
| | Independent | *E*-taen Thai farm tractors | Tran_ETN_L | Natural logarithm of total number of *E*-taen Thai farm tractors | Demand | CDD |
| | Independent | Cars | Tran_Car_L | Natural logarithm of total number of cars | Demand | CDD |
| Disaster | Independent | Households affected by natural disasters | DisHH | Total number of households affected by natural disasters in previous year | Demand | CDD |
| | Independent | Risk of natural disasters | DisRisk | Whether sub-district is at risk of natural disasters | Demand | CDD |

**Table 2**
Descriptive statistics of variables.

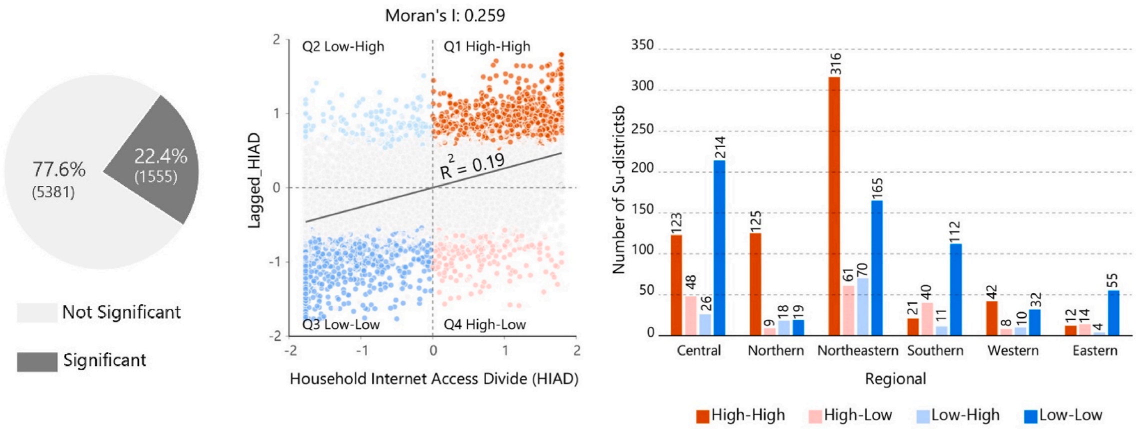
| Variable | Mean | Median | Maximum | Q.25 | Q.75 | SD | Skew | Kurtosis | VIF |
|---|---|---|---|---|---|---|---|---|---|
| HIAD | 0.491 | 0.512 | 1.000 | 0.267 | 0.719 | 0.282 | −0.135 | −1.043 | – |
| PopDens_L | 4.320 | 4.310 | 11.180 | 3.640 | 4.870 | 1.170 | 1.040 | 4.280 | 1.087 |
| IncFarm_L | 10.760 | 11.210 | 18.210 | 10.700 | 11.730 | 2.510 | −3.430 | 12.390 | 1.567 |
| IncNFarm_L | 9.860 | 10.920 | 17.950 | 10.070 | 11.540 | 3.480 | −2.180 | 3.640 | 1.589 |
| EngPro | 0.270 | 0.210 | 1.000 | 0.070 | 0.410 | 0.250 | 1.010 | 0.290 | 1.061 |
| Gov_ICT_L | 3.230 | 3.710 | 9.650 | 1.610 | 4.790 | 2.060 | −0.320 | −0.950 | 1.104 |
| SocActv | 0.760 | 0.830 | 1.000 | 0.640 | 0.950 | 0.240 | −1.180 | 0.790 | 1.081 |
| SocGame_L | 0.690 | 0.000 | 4.250 | 0.000 | 1.100 | 0.800 | 0.760 | −0.380 | 1.115 |
| Tran_MTB_L | 7.160 | 7.210 | 10.960 | 6.840 | 7.550 | 0.650 | −1.770 | 14.730 | 2.964 |
| Tran_ETN_L | 3.360 | 3.760 | 9.250 | 2.080 | 4.860 | 1.930 | −0.450 | −0.840 | 1.125 |
| Tran_Car_L | 6.570 | 6.610 | 11.020 | 6.170 | 7.050 | 0.820 | −1.530 | 10.620 | 2.822 |
| DisHH | 0.599 | 1.000 | 1.000 | 1.000 | 1.000 | 0.490 | −0.400 | −1.840 | 1.134 |
| DisRisk | 0.010 | 0.000 | 1.000 | 0.000 | 0.000 | 0.060 | 9.210 | 109.140 | 1.029 |

Note: VIF = variance inflation factor; SD represents standard deviation.

the ratio of households without an Internet connection to the sum of households at the sub-district level, served as the primary indicator. In this study, the HIAD is operationalized inversely to household internet usage; specifically, a higher HIAD level indicates a lower level of internet use among households, which signifies a greater digital divide within the sub-district. Conversely, a lower HIAD level indicates a higher level of internet use among households, signifying a smaller digital divide within the sub-district.

The first result of the global Moran's *I* analysis with a positive spatial autocorrelation of 0.259 and a *p*-value less than 0.00 rejected the null hypothesis that the pattern of the HIAD was randomly distributed. This result confirms the presence of the spatial clustering of the HIAD at the sub-district level across the country, suggesting that neighboring sub-districts tend to have similar levels of household internet access or non-access. LISA was subsequently utilized to identify and evaluate the nature of local spatial clustering. As shown in Fig. 2a, about 22.4% of sub-districts exhibit significant clustering, implying a systematic spatial variation in the HIAD. The LISA scatterplot, divided into four quadrants, represents the different types of spatial associations: high-high (Q1), low-high (Q2), low-low (Q3), and high-low (Q4). In Fig. 2b, the scatterplot highlights a notable concentration of sub-districts in the high-high and low-low quadrants, Q1 and Q3, respectively. These clusters indicate regions where sub-districts with high HIAD are surrounded by sub-districts with similarly high HIAD values, and vice versa for low HIAD values, suggesting areas with uniformly distributed internet connectivity issues or uniformly better connectivity. Moreover, Fig. 2c delineates the regional breakdown of these clustering types. It illustrates that certain region, such as the northeastern and northern parts of Thailand, predominantly exhibit high-high clustering, indicating extensive areas with low levels of household internet usage. Conversely, the central region, including the Bangkok metropolitan areas, shows a predominance of low-low clusters, which suggests higher internet usage among households.

This geographical distribution of HIAD patterns is indicative of a digital divide that is not randomly dispersed but rather systematically structured across the country's sub-districts. The high concentration of high-HIAD clusters in the northern and




(a) (b) (c)

**Fig. 2.** Number of sub-districts and percentages in accordance with Moran's *I* LISA spatial statistics (a), the Moran's *I* scatterplot diagram of the HIAD (b), and LISA clustering types categorized by region (c).

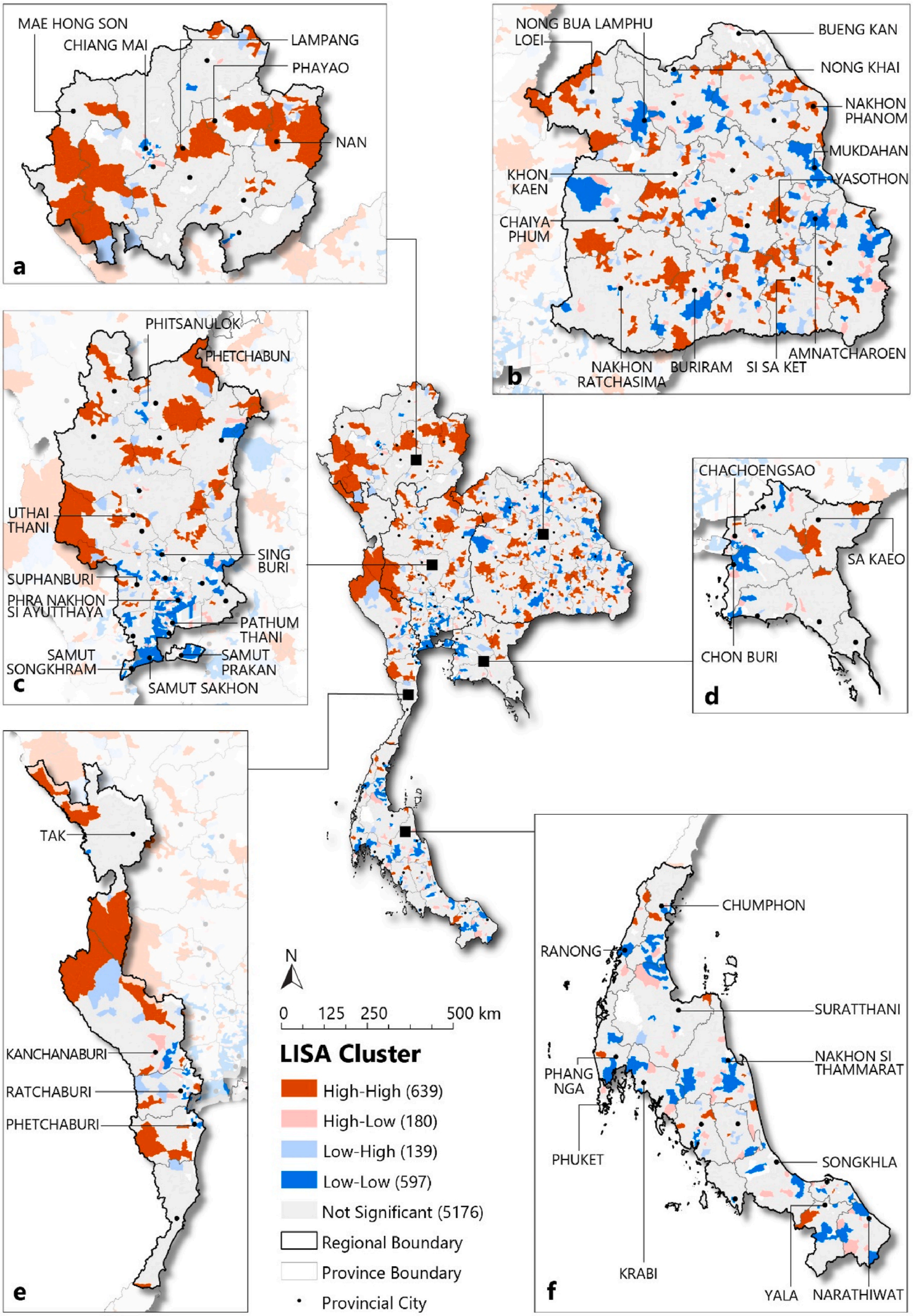
northeastern regions could reflect regional disparities in digital infrastructure development, socioeconomic status, and access to digital literacy resources. On the other hand, the central region's low-low clusters might be benefiting from the proximity to urban centers, where there's often greater investment in connectivity infrastructure and digital services.

The LISA cluster map illustrated in Fig. 3 provides a nuanced visualization of the digital divide as measured by the HIAD across Thailand's diverse geographic regions. Overall, the comprehensive analysis of the LISA reveals a multifaceted pattern of the digital divide across Thailand's sub-districts, shedding light on both the geographic and socioeconomic contours of internet connectivity in the country. The results revealed a high–high clustering type refers to those sub-districts with a high HIAD value surrounded by similar features. The majority of high–high clustering types are located in the northeastern region, followed by the northern and central regions, respectively. The location patterns of high–high clusters vary by region. A scattered pattern across the region appears in the northeastern and central regions. The majority of high–high clusters in the northern and western regions are agglomerated densely along the country's border. Only a small number of high–high clusters appear in the eastern and southern regions. Meanwhile, a low–low clustering type means a sub-district with low HIAD value surrounded by other sub-districts with similar HIAD values. The co-agglomeration of subdistricts and their adjacent neighbors with a low HIAD value is clearly seen in the central region, followed by the northern and southern regions. Bangkok, the capital city, is surrounded by a dense agglomeration of low–low clusters in the central and eastern regions. These regions contain a large urban agglomeration that comprises the center of economic activity in Thailand, reflecting the locations of company headquarters and many manufacturers. The minority of low–low cluster agglomeration can be seen in the southern region, which contains Phang Nga, Krabi, and Nakhon Si Thammarat: provincial areas with vibrant tourism industry activities.

Moreover, the LISA results revealed the significant location pattern of spatial outliers with a negative spatial correlation. The clusters of spatial outliers are more fragmented than those with positive spatial correlations. It is noteworthy that their location seems to appear directly next to the opposite pole of Internet access value. A low–high cluster is often found next to a high–high cluster. For example, a pattern of spatial outliers in the northern and western regions shows low–high clusters located next to high–high clusters in Mae Hong Son and Kanchanaburi provinces. In the same vein, low–low clusters in the southern region are surrounded by a scatter pattern of high–low clusters, as seen in Phang Nga, Krabi, and Nakhon Si Thammarat. For the northeastern region, although it has the highest number of high–high clusters, it also captures a diverse combination of clustering types. The results show a distinctive pattern between the upper and lower halves of the region. A low–low cluster together with a high–low cluster can be found in the upper half of the region, such as in Nong Bua Lamphu, Udon Thani, and Maha Sarakham. Meanwhile, the lower part contains a group of a high–high cluster and a low–high cluster, as seen in Nakhon Rachasima, Buriram, Surin, and Sisaket. Despite the majority of positive spatial autocorrelations, many significant outliers provide evidence of the spatial disparities at the regional scale, which raises questions about the absence of regional development manifestation.

### 5.2. *Spatial econometric models*

In this section, spatial regression models were used to determine the most influential exploratory variables in various socioeconomic components of the HIAD. The study took place at the country level. Table 3 summarizes the maximum likelihood estimation results of the formulated SLM and SEM. The queen contiguity weight was used as the spatial weighting method. We first conducted a Lagrange multiplier (LM) test to determine whether an SLM or SEM model was more appropriate for estimation. The LM test oriented us toward an SLM, where the robust LM error failed to reject the hypotheses. The LM test results are provided in Appendix A. We also




(caption on next page)

**Fig. 3.** A LISA cluster analysis of Thailand; (a) northern region, (b) northeastern region, (c) northeastern region, (d) eastern region, (e) western region, and (f) southern region.

estimated the SEM as being capable of the most robust analysis.

As shown in Table 3, the SLM and SEM are both significantly outperformed OLS. Both autoregressive lag coefficients of lambda and rho were found to be strongly significant ($p < 0.01$). Moreover, the results for the Akaike information criterion (AIC) confirmed that the SLM had the best goodness of fit of the models (AIC = 686.84), followed by the SEM (AIC = 805.95) and OLS (AIC = 1341.70).

Overall, the results for the country-wide model are shown in Table 3. All explanatory variables were significantly estimated at the 99% level ($p < 0.01$). All three models (i.e., OLS, SLM, and SEM) suggested a strongest positive association between the total number of households affected by natural disasters (*DisHH*) and the HIAD. Similarly, sub-districts with natural disaster risks (*DisRisk*) had significant spatial positive associations with the HIAD. That is, the HIAD is prominent in the most disaster-prone areas. This finding is in conflict with the emerging trend of using telecommunications technologies to enhance disaster risk mitigation while Internet access is significantly unequal. Alongside this, we also found a spatial positive association of the HIAD with the number of farming trucks (*Tran_ETN_L*), total number of motorcycles (*Tran_MTB_L*), and income per farming household (*IncFarm_L*). Income per non-farming household (*IncNFarm_L*) shows a spatial negative coefficient with the HIAD. Population density (*PopDens_L*) and total number of cars (*Tran_Car_L*) were positively correlated with the HIAD, whereas coefficients were negative for the highly skilled population cover variables, namely individuals with English proficiency (*EngPro*) and number of individuals with ICT training participation (*Gov_ICT_L*). This indicates the importance of English skills and government support in improving the digital divide at the sub-district level in Thailand.

Specifically, these empirical results provided robust support for our proposed hypotheses regarding the determinants of HIAD. First, Hypothesis 1 (H1) posited a negative relationship between population density and HIAD. The empirical data supported this hypothesis, indicating a significant negative relationship ($\beta = -0.019$, $p < 0.01$). This suggests that higher population density is associated with lower HIAD, meaning more densely populated sub-districts tend to have better internet access. Second, Hypothesis 2 (H2) suggested a negative relationship between the income of farming households and HIAD. This hypothesis was supported, as the results showed a significant negative relationship ($\beta = -0.004$, $p < 0.01$). Higher income in farming households is linked to a wider HIAD, indicating poorer internet access. In addition, Hypothesis 3 (H3) proposed a negative relationship between the income of non-

**Table 3**
Estimation results of spatial econometric models at country level.

| Variable | OLS | SLM | SEM |
|---|---|---|---|
| PopDens_L | −0.043*** | −0.019*** | −0.036*** |
| | (-0.003) | (-0.003) | (-0.003) |
| IncFarm_L | 0.006*** | 0.004*** | 0.004*** |
| | (-0.002) | (-0.001) | (-0.002) |
| IncNFarm_L | −0.005*** | −0.004*** | −0.003*** |
| | (-0.001) | (-0.001) | (-0.001) |
| EngPro | −0.079*** | −0.066*** | −0.066*** |
| | (-0.013) | (-0.012) | (-0.013) |
| GOV_ICT_L | −0.008*** | −0.006*** | −0.006*** |
| | (-0.002) | (-0.002) | (-0.002) |
| SocActv | −0.047*** | −0.053*** | −0.058*** |
| | (-0.014) | (-0.013) | (-0.014) |
| SocGame_L | 0.013*** | 0.013*** | 0.014*** |
| | (-0.004) | (-0.004) | (-0.004) |
| Tran_MTB_L | 0.041*** | 0.038*** | 0.028*** |
| | (-0.008) | (-0.008) | (-0.009) |
| Tran_ETN_L | 0.015*** | 0.008*** | 0.010*** |
| | (-0.002) | (-0.002) | (-0.002) |
| Tran_Car_L | −0.072*** | −0.070*** | −0.055*** |
| | (-0.007) | (-0.006) | (-0.007) |
| DisHH | 0.200*** | 0.169*** | 0.218*** |
| | (-0.050) | (-0.047) | (-0.051) |
| DisRisk | 0.046*** | 0.039*** | 0.043*** |
| | (-0.007) | (-0.007) | (-0.007) |
| Constant | 0.837*** | 0.575*** | 0.802*** |
| | (-0.039) | (-0.038) | (-0.04) |
| Observations | 6936 | 6936 | 6936 |
| $R^2$ | 0.108 | | |
| Adjusted $R^2$ | 0.107 | | |
| Pseudo-$R^2$ | | 0.214 | 0.200 |
| AIC | | 686.838 | 805.95 |
| BIC | | 789.505 | 908.617 |
| Lambda | | | 0.398*** |
| Rho | | 0.397*** | |

Note: *** = $p < 0.01$, ** = $p < 0.05$; * = $p < 0.1$. The standard error is presented in parentheses.

farming households and HIAD. The empirical data supported this hypothesis, showing a significant negative relationship ($\beta = -0.004$, $p < 0.01$), suggesting that higher income in non-farming households correlates with better internet access and lower HIAD. Moreover, Hypothesis 4 (H4) posited a negative relationship between English proficiency and HIAD. The results supported this hypothesis, revealing a significant negative relationship ($\beta = -0.066, p < 0.01$). Higher English proficiency within a sub-district is associated with better internet access and lower HIAD. Furthermore, Hypothesis 5 (H5) suggested a negative relationship between participation in government-initiated ICT training and HIAD. The hypothesis was supported, with a significant negative relationship found ($\beta = -0.006, p < 0.01$). Greater participation in ICT training programs is linked to better internet access and lower HIAD. Next, Hypothesis 6 (H6) proposed a negative relationship between social activity participation and HIAD. The results supported this hypothesis, showing a significant negative relationship ($\beta = -0.053$, $p < 0.01$). Higher levels of social activity participation are associated with better internet access and lower HIAD. In contrast, Hypothesis 7 (H7) posited a positive relationship between the density of game cafés and HIAD. This hypothesis was supported, with the empirical data showing a significant positive relationship ($\beta = 0.013$, $p < 0.01$). A higher density of game cafés is correlated with higher HIAD, indicating poorer internet access. Similarly, Hypothesis 8 (H8) suggested a positive relationship between motorcycle ownership and HIAD. The results supported this hypothesis, revealing a significant positive relationship ($\beta = 0.038, p < 0.01$). Greater motorcycle ownership is linked to higher HIAD, indicating poorer internet access. Further, Hypothesis 9 (H9) proposed a positive relationship between *E*-taen tractor ownership and HIAD. The hypothesis was supported, with the empirical data showing a significant positive relationship ($\beta = 0.008, p < 0.01$). Higher *E*-taen tractor ownership is associated with a higher HIAD, indicating poorer internet access. Additionally, Hypothesis 10 (H10) posited a negative relationship between car ownership and HIAD. This hypothesis was supported, with a significant negative relationship found ($\beta = -0.070, p < 0.01$). Higher car ownership is linked to better internet access and lower HIAD. Hypothesis 11 (H11) suggested a positive relationship between the number of households affected by natural disasters and HIAD. The results supported this hypothesis, revealing a significant positive relationship ($\beta = 0.169, p < 0.01$). More households affected by natural disasters are associated with higher HIAD, indicating poorer internet access. Lastly, Hypothesis 12 (H12) also proposed a positive relationship between the risk of natural disasters and HIAD. This hypothesis was supported, with the empirical data showing a significant positive relationship ($\beta = 0.039, p < 0.01$). Sub-districts with higher risk of natural disasters are correlated with higher HIAD, indicating poorer internet access.

Therefore, we conclude that HIAD is at low levels for sub-districts with dense urbanization, high incomes, high English comprehension capability and ICT competencies, high levels of social activity participation, and high levels of car ownership. Conversely, sub-districts with a higher density of game cafés, greater motorcycle ownership, more *E*-taen tractors, more households affected by natural disasters, and higher risk of natural disasters tend to have higher HIAD, indicating poorer internet access.

## 6. Conclusions and implications

This study aims to explore the spatial distribution of the digital divide and its determinants at a sub-district level in Thailand using a dataset of information on basic household necessities from 2021, which was provided by the Community Development Department of Thailand. The results of the spatial autocorrelation testing reveal a significant positive spatial agglomeration of the digital divide at a sub-district level in Thailand. LISA scatterplots illustrated that the spatial characteristics of the digital divide (i.e., HIAD) tended to be high–high in sub-districts in the northeastern and northern regions of Thailand. This implies that sub-districts with high HIAD values were surrounded by sub-districts with similar HIAD levels. Furthermore, these two regions have the highest poverty rates in Thailand (National Statistical Office of Thailand, 2018) and the lowest gross regional product per capita (GRP; National Economic and Social Development Board, 2018). Government support in terms of providing household internet access is required in struggling areas. The results also revealed low–low agglomerations in the spatial distribution of HIAD at a sub-district level in Thailand's central region, which has high levels of economic activity and greater urbanization. Based on these findings, we identified the existence of deeper spatial disparities in HIAD than have been found in previous studies (e.g., Pick et al., 2021; Song et al., 2020; Szeles, 2018) that did not analyze the digital divide on a smaller scale, below the level of the prefecture or province. This study showed mixed spatial patterns of HIAD within certain provinces that suggest that local and central authorities must aim to narrow the digital divide. Inequalities in socioeconomic development, even within the same province, could widen the digital divide in terms of HIAD. According to *Thailand's Poverty and Inequality Report 2020* published by the Office of the National Economic and Social Development Council of Thailand (2021), 45.37% of poor households can access the internet, meaning that more than half of poor households still do not have access to the internet. This is a large proportion compared to wealthy households without internet access. Accordingly, the government must extend support through the provision of computer equipment, free internet access, and ICT training programs, particularly targeting poor households in remote areas outside municipalities. Specifically, the government should implement a targeted digital inclusion strategy focusing on the most affected sub-districts, especially those in the northeastern and northern regions of Thailand. These efforts align with and could enhance the effectiveness of the Digital Economy and Society Development Plan, the Digital Economy Promotion Master Plan, and the Thailand 4.0 initiative. Furthermore, these findings underscore the need for a decentralized approach to digital policy in Thailand, as a homogeneous policy may prove ineffective due to the variations in the digital divide across the country. Prioritizing local contexts would significantly bolster the effectiveness of strategies aimed at bridging this divide.

In terms of the spatial regression analysis, the results show that all of the independent variable categories proposed in the conceptual model were significantly associated with the HIAD. Specifically, it was revealed that the average annual income per farming household, the number of game cafés, the number of motorcycles, the number of *E*-Taen Thai farm tractors, the number of households affected by natural disasters, and the risk of natural disasters are all positively correlated with HIAD. In addition, we found that HIAD is negatively affected by the population density, the average annual income per non-farming household, the average English proficiency, the degree of participation in government-initiated ICT training, the degree of social activity participation, and the number of cars. Although the results are mixed in some variable categories, this analysis nonetheless provides new insights into the relationships between these variables and HIAD.

Consistent with our results, several studies have argued that economic conditions can increase the likelihood of individuals having internet access (e.g., Pick et al., 2021; Setthasuravich & Kato, 2020, 2022; Song et al., 2020; Szeles, 2018). However, our results also provide further insights; specifically, we found that a negative relationship between the income of farming households and HIAD in Thai sub-districts. This indicates that farming households are more likely to have difficulty accessing the internet at their houses. One possible explanation is that even farming communities with higher incomes continue to face digital divides, possibly due to lower digital literacy levels, perceptions of internet services as irrelevant, and constraints in infrastructure in rural areas (Wims, 2014). Furthermore, among households in the agricultural sector, trends in access to the internet could be led by broader changes in agriculture-related technology adoption (Zheng et al., 2022). This finding supports the smart farming policy and indicates that governments should support internet use in farming households to promote agricultural livelihoods and the adoption of ICT in this sector (Nguyen et al., 2022), particularly by offering digital literacy programs providing necessary information about the benefits and concerns of internet connectivity. This finding also supports the Net Pracharat Project by identifying specific sub-districts to which the Thai government can provide high-speed internet access.

In contrast, the findings also revealed a positive relationship between the income of non-farming households and HIAD. This suggests that non-farming household incomes are often directly connected to sectors like services and manufacturing that benefit from or require digital connectivity (Mehta, 2017; Wang et al., 2023). The observed relationship could therefore be explained by the fact that non-farming sectors typically involve greater use of digital technology compared to agricultural sectors. This technological integration encourages the acquisition of digital skills and resources. For instance, jobs in the service sector might require employees to interact with digital platforms for tasks ranging from scheduling and customer service to remote work capabilities. Similarly, manufacturing sectors are incorporating digital technologies for production management and supply chain logistics. These sectors not only provide employees with the economic means to obtain household internet access but also demand digital literacy among workers, encouraging households to invest in better internet infrastructure. Furthermore, the skills acquired in these jobs can increase the perceived relevance and utility of the internet, unlike in farming communities. To support the growing demand for digital connectivity in non-farming households, policymakers should focus on improving internet infrastructure, especially in urban and semi-urban areas where these sectors are concentrated. This includes not only the expansion of broadband access but also the enhancement of internet speed and reliability.

In the social capital category, the social activity participation variable was seen to have a negative relationship with the HIAD. The likelihood of internet usage is increased in households that are more closely connected with their communities and have high levels of participation in social activities. This effect is primarily due to the increased social interactions that encourage the sharing and dissemination of technological knowledge and resources. Engaging in community activities fosters environments where individuals feel supported in exploring new technologies, which in turn can lead to higher rates of digital adoption (Ludwig et al., 2019; Pederzoli, 2019, pp. 395–408). This finding supports the "Project to Upgrade the Community ICT Learning Center to a Community Digital Center, " which is a part of the Digital Economy and Society Development Plan. Given the positive impact of social activity participation on reducing the HIAD, it would be beneficial to promote and invest in community digital centers that offer digital training and resources. These centers can serve as hubs for community engagement and digital education, creating an environment that encourages and facilitates the adoption of internet technology by individuals across different demographic groups. In contrast, the number of game cafés in each sub-district is positively correlated with HIAD, meaning that this variable may widen the HIAD. While these establishments are indeed social hubs that promote digital engagement and learning through peer interaction, higher numbers of game cafés may reflect inadequate internet infrastructure and a lack of adequate home-based internet access in certain areas (Wongyai & Huttasin, 2022). As this finding indicates a potential deficiency in home-based internet infrastructure, one policy implication is that the government should support the role of game cafés as community digital hubs while simultaneously improving home-based internet access. This would help to ensure that these cafés complement rather than substitute for comprehensive home internet access.

As anticipated, we also found that population density, average English proficiency, and engagement with government-initiated ICT training can narrow the HIAD. These results are consistent with those of other studies (Nishida et al., 2014; Pick & Azari, 2011; Pick et al., 2015) and suggest that demographics, education, and ICT training programs are associated with the digital divide.

The results revealed a significant negative relationship between population density and the HIAD. This suggests that government policies and initiatives related to ICT development may differ by region according to population density, typically favoring regions with a more concentrated population that are perceived to be more economically viable. Another possible explanation for this effect is that areas with higher population density often experience greater demand and more robust economic activities. Consequently, these regions are likely to possess more developed ICT infrastructure compared to areas with lower population density. One practical policy implication suggested by this findings is that the government should implement targeted ICT infrastructure development in areas with low population density. This approach would aim to balance the digital infrastructure across various regions, ensuring that rural and less populated areas are not left behind in terms of digital advancement. This finding provides additional support for the Net Pracharat Project, which offers people in rural and less populated areas the chance to access high-speed internet for free.

The results also unveiled a negative association between average English proficiency and HIAD. This relationship could be explained that English is often considered the lingua franca of the internet, with a significant portion of web content, especially technical and educational resources, available predominantly in English. Greater English proficiency enables individuals to access, understand, and utilize these resources more effectively, thus reducing the digital divide. Another possible explanation may come from a supply-side perspective. This negative association can be interpreted as reflecting the educational infrastructure in specific areas.

High levels of English proficiency in a region might indicate a well-developed educational infrastructure that not only emphasizes language skills but also integrates digital literacy into the curriculum. Schools and educational institutions that offer strong English language programs often have better overall resources, including greater access to computers and the internet, which contributes to digital skills development (Ahmed & Roche, 2021; Oyedemi & Mogano, 2018). Based on this finding, the government should integrate comprehensive digital literacy training into English language education programs, especially in areas with lower digital access. This policy would aim to use English language classes as a platform for enhancing digital skills. The government may consider allocating increased funding and resources to schools and educational institutions to support the dual delivery of English and digital literacy education. This could include investments in computer labs, digital learning tools, and high-speed internet infrastructure in educational settings.

Moreover, our results demonstrated a significant negative correlation between participation in government-initiated ICT training and HIAD. This suggests that sub-districts with higher participation in ICT training are likely to experience lower levels of HIAD. Government-initiated ICT training programs often come with investments in ICT infrastructure to support the training efforts. This includes upgrading or installing new internet facilities and providing the necessary hardware and software in training centers and local communities, thereby facilitating broader internet access and usage in households. As a result, areas with higher participation in these programs are likely to benefit from improved digital infrastructure. This finding supports the Thai government's efforts to provide ICT programs to the local community through the work of Digital Volunteers in community digital centers. The government should expand and diversify ICT training programs, making them available in more geographic regions and particularly focusing on areas with currently high HIAD levels.

This study also found that the number of motorcycles, cars, and *E*-Taen tractors is related to HIAD. We found that motorcycle ownership has a significant positive effect on the HIAD. This indicates that sub-districts with higher levels of motorcycle ownership are likely to exhibit greater levels of HIAD. One possible explanation of this effect is that households might allocate a substantial part of their limited financial resources to purchasing motorcycles, potentially at the cost of other non-essential services such as household internet connectivity. This trade-off is particularly prevalent in less affluent areas where the necessity of transportation for work or education takes precedence over the perceived need for internet access (Forman et al., 2021, pp. 31–45). This finding could suggest that the government should implement targeted financial assistance programs that help balance transportation and internet connectivity needs. This initiative would aim to alleviate the financial burden of transportation costs, freeing up resources that could then be directed toward improving household internet access. We also found that the number of *E*-Taen Thai farm tractors is positively correlated with HIAD. This indicator is closely linked to individuals' economic backgrounds and occupations (i.e., working in the agricultural sector), which may affect individuals' opportunities to access the digital world. The considerable expenses associated with purchasing and maintaining a tractor can exhaust financial resources that might otherwise be used for technological improvements, including internet services. In such scenarios, technological investment is often focused on enhancing agricultural productivity rather than on expanding digital connectivity. Based on this result, the government should implement subsidy programs specifically designed to support technology access and internet connectivity for agricultural workers. This would help to offset the financial burden of agricultural investments, enabling agricultural workers to also invest in digital technologies. Ultimately, this would contribute to the smart farming policy. Additionally, our findings revealed a positive relationship between car ownership and HIAD. This indicates that sub-districts with higher levels of car ownership are likely to exhibit lower levels of HIAD. Increased car ownership in households may improve individuals' mobility, thus increasing their economic activity and potentially motivating them to ensure internet access at their homes to support such activities. The study results indicate that sub-districts with high-quality road infrastructure can facilitate increased car ownership, which in turn may also allow internet providers to provide services more readily in these areas. One key policy implication arising from this result is that the government should allocate significant government funding to improving and expanding road networks, especially in sub-districts with lower car ownership and higher HIAD. Better roads can enhance accessibility, making it easier for residents to commute and for internet service providers to install and maintain network infrastructure. These factors have not been found to be correlated with HIAD in any previous studies. This finding thus provides a new variable for spatial studies of the digital divide in developing countries.

Our results also revealed that disaster category variables are correlated with the HIAD—to the best of our knowledge, this is the first study that has identified such a relationship. The HIAD was found to be related to the number of natural disasters in a sub-district or and was higher in areas at greater risk. This effect is unsurprising since areas more susceptible to natural disasters and those recently impacted by such events often face greater challenges in accessing the internet. Natural disasters can damage or destroy infrastructure, including communication networks, hampering internet connectivity. Furthermore, in the aftermath of disasters, the focus and resources of both the community and the government are typically redirected from digital infrastructure development to urgent relief and recovery efforts. In other words, the HIAD is a crucial problem in the most disaster-prone areas. These findings support the national disaster management policy, which mandates that when natural disasters occur, governments should ensure that all people and households are informed within a short time after the disaster. Providing improved ICT infrastructure coverage in disaster-prone areas is thus a highly important issue for governments in developing countries to consider.

In summary, these policy implications previously detailed are organized into two primary categories: *Availability of ICT Services* and *Capacity of ICT Usages*, each addressing different aspects of the digital divide in Thailand. First, *Availability of ICT Services* focuses on enhancing the physical infrastructure and access to internet services across diverse regions. This includes extending government support through the provision of computer equipment and free internet access, targeting impoverished households in remote areas, particularly in the Northeastern and Northern regions of Thailand. These efforts are in line with the Digital Economy and Society Development Plan and the Thailand 4.0 initiative. Additional recommendations include the expansion of broadband access, speed, and reliability, especially in urban and semi-urban areas where demand is concentrated. Furthermore, improving ICT infrastructure coverage in disaster-prone areas is vital, supporting national disaster management policies to ensure rapid communication post-disaster. Suggestions also extend to significant government funding for improving and expanding road networks, which can enhance accessibility and facilitate the installation and maintenance of network infrastructure in areas with lower car ownership. Second, *Capacity of ICT Usages* emphasizes educational and community-based programs to enhance the effective use of technology. Key proposals include integrating digital literacy training into English language education programs and enhancing community digital centers. These centers, promoted as hubs for community engagement and digital education, are particularly beneficial in regions with high HIAD levels. Expanding ICT training programs through Digital Volunteers and targeted digital inclusion strategies are recommended. These strategies should include subsidies for technology access for agricultural workers to support the smart farming policy and implementing targeted financial assistance programs that help balance transportation and internet connectivity needs, potentially freeing up household resources for internet access. Furthermore, supporting internet use in farming households through digital literacy programs and enhancing non-farming household digital connectivity demand cater to specific demographic and occupational needs within the community.

Based on the findings and policy implications proposed above, this study complements the existing literature on the digital divide by providing empirical evidence of the spatial patterns of the digital divide at a sub-district level in Thailand. We expect that the new variables established can be used to complement the Spatially Aware Technology Utilization Model (SATUM). However, these findings are limited by the use of a one-year dataset that could not illustrate the temporal development of the observed spatial patterns in HIAD. Using cross-sectional data also limits the potential to establish causality between the observed digital divide and its various determinants. Moreover, our findings are partly constrained by the availability of data, particularly concerning the quality of internet infrastructure, the availability of service providers, and other related supply-side factors. Despite these limitations, investigations relying on demand-side variables still provide valuable insights into the digital divide across Thailand's sub-districts. Furthermore, while the study includes a wide range of variables, it cannot account for all possible factors influencing the digital divide, such as cultural influences and individual-level data on technology adoption rates. Although the data were detailed, the granularity may still conceal disparities within sub-districts that are small in scale but significant in impact. Additionally, the reliance on official government data for this analysis may introduce biases if there are inaccuracies or inconsistencies in data collection or reporting methodologies across different sub-districts.

Therefore, future research directions should consider longitudinal studies incorporating multi-year datasets to elucidate trends and changes in the digital divide's spatial patterns over time. Such studies could enhance the understanding of causal relationships between the digital divide and its determinants. Expanding the scope of research to include additional variables, including critical supply-side factors such as internet infrastructure quality and service provider availability, alongside the influence of cultural dynamics and the effectiveness of digital literacy programs, could provide a more holistic view of the factors contributing to the digital divide. Investigations at a micro-level, using household or individual data, would offer insights into more nuanced patterns and determinants of the digital divide, potentially uncovering localized disparities within sub-districts. Finally, comparative analyses with other countries or regions could place Thailand's digital divide within a broader context of global digital inclusion efforts and policy interventions. These future studies would not only address the limitations of the current research but also contribute valuable knowledge to the ongoing efforts to bridge the digital divide.

## CRediT authorship contribution statement

**Prasongchai Setthasuravich:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kulacha Sirikhan:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Hironori Kato:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

## Declaration of competing interest

None.

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## Appendix

**Appendix A**
Lagrange multiplier test of spatial dependence

| Diagnostic for spatial dependence | Value | *p* |
|---|---|---|
| Lagrange multiplier (LM; lag) | 772.971 | <0.00 |
| Robust LM (lag) | 150.317 | <0.00 |
| LM (error) | 623.436 | <0.00 |
| Robust LM (error) | 0.782 | 0.38 |
| LM (SARMA) | 773.753 | <0.00 |

## References

Ahmed, S. T., & Roche, T. (2021). Making the connection: Examining the relationship between undergraduate students' digital literacy and academic success in an English medium instruction (EMI) university. *Education and Information Technologies, 26*(4), 4601–4620. https://doi.org/10.1007/s10639-021-10443-0

Ali, K. S., & Faroque, A. R. (2023). Addressing the complexity of the digital divide and the role of government in addressing it: Role of government in bridging the digital divide. In F. Sustainable (Ed.), *Businesses in emerging economies* (pp. 127–145). Emerald Publishing Limited. https://doi.org/10.1108/978-1-80455-640-520231009.

Alsmari, N. A. (2021). *The relationship between language proficiency level and L2*.

Anselin, L. (2002). Under the hood Issues in the specification and interpretation of spatial regression models. *Agricultural Economics, 27*(3), 247–267. https://doi.org/10.1111/j.1574-0862.2002.tb00120.x

Anselin, L., Syabri, I., & Kho, Y. (2010). GeoDa: An introduction to spatial data analysis. In M. Fischer, & A. Getis (Eds.), *Handbook of applied spatial analysis*. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-642-03647-7_5.

Armey, L. E., & Hosman, L. (2016). The centrality of electricity to ICT use in low-income countries. *Telecommunications Policy, 40*(7), 617–627. https://doi.org/10.1016/j.telpol.2015.08.005

Ashraf, T., Kadir, S. M. M., & Saha, S. (2015). *Development of a context-appropriate*.

Asia-Pacific Telecommunity. (2019). Village broadband internet project (Net Pracharat) of Thailand. *Ministry of Digital Economy and Society*. https://npcradm.netpracharat.com//mediadetail/qbHuL1RqXN.pdf.

Asrani, C., & Kar, A. K. (2022). Diffusion and adoption of digital communications services in India. *Information Technology for Development*, 1–23. https://doi.org/10.1080/02681102.2022.2046536

Bánhidi, Z. (2021). The impact of broadband networks on growth and development in South America. *Periodica Polytechnica, Social and Management Sciences, 29*(1), 33–39. https://doi.org/10.3311/PPso.14905

Bahrini, R., & Qaffas, A. A. (2019). Impact of information and communication technology on economic growth: Evidence from developing countries. *Economies, 7*(1), 21. https://doi.org/10.3390/economies7010021

Bangkok Post. (2021a). Govt relief causes 'havoc'. *[online]*. Available at: https://www.bangkokpost.com/opinion/opinion/2069519/govt-relief-causes-havoc. (Accessed 14 December 2021).

Bangkok Post. (2021b). Not ready for online classes [online]. Available at https://www.bangkokpost.com/opinion/opinion/2049359/not-ready-for-online-classes. (Accessed 14 December 2021).

Barbier, E. B. (2023). Overcoming digital poverty traps in rural Asia. *Review of Development Economics, 27*(3), 1403–1420. https://doi.org/10.1111/rode.12962

Barry, J. J. (2018). *Information communication technology and poverty alleviation: Promoting good governance in the developing world*. Routledge. https://doi.org/10.4324/9780429504204

Bhui, S., & Pandit, D. (2021). Underlying factors and dependencies towards a dynamic vehicle ownership model in India: A content analysis approach. In *Conference of transportation research group of India* (pp. 155–175). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-19-3494-0_10.

Billon, M., Crespo, J., & Lera-Lopez, F. (2021). Do educational inequalities affect internet use? An analysis for developed and developing countries. *Telematics and Informatics, 58*, Article 101521. https://doi.org/10.1016/j.tele.2020.101521

Chang, Y., Kim, H., Wong, S. F., & Park, M. C. (2015). A comparison of the digital divide across three countries with different development indices. *Journal of Global Information Management, 23*(4), 55–76. https://doi.org/10.4018/JGIM.2015100103

Chiu, B. Y. (2023). Relationship between motorcycle travel and the built environment: Evidence from Taipei, Taiwan. *Journal of Transport Geography, 110*, Article 103607. https://doi.org/10.1016/j.jtrangeo.2023.103607

Chohan, S. R., & Hu, G. (2022). Strengthening digital inclusion through e-government: Cohesive ICT training programs to intensify digital competency. *Information Technology for Development, 28*(1), 16–38. https://doi.org/10.1080/02681102.2020.1841713

Dargin, J. S., Fan, C., & Mostafavi, A. (2021). Vulnerable populations and social media use in disasters: Uncovering the digital divide in three major US hurricanes. *International Journal of Disaster Risk Reduction, 54*, Article 102043. https://doi.org/10.1016/j.ijdrr.2021.102043

Dkhil, I. B., & Jebsi, K. (2020). Access regulation and broadband deployment: Evidence from a worldwide dataset. *International Journal of the Economics of Business, 27*(2), 203–253.

Durand, A., Zijlstra, T., van Oort, N., Hoogendoorn-Lanser, S., & Hoogendoorn, S. (2022). Access denied? Digital inequality in transport services. *Transport Reviews, 42*(1), 32. https://doi.org/10.1080/01441647.2021.1923584

Elkhatib, Y., Tyson, G., & Sathiaseelan, A. (2015). Does the Internet deserve everybody?. In *Proceedings of the 2015 ACM SIGCOMM workshop on ethics in networked systems research* (pp. 5–8).

Fakhruddin, S. H. M. (2024). Disasters in Thailand. In *Handbook of disaster risk reduction & management*. World Scientific Publishing. https://doi.org/10.1142/9789813207950_0027 (Chapter 27)).

Farhadi, M., & Ismail, R. (2014). The impact of information and communication technology availability on economic growth. *Research Journal of Applied Sciences, Engineering and Technology, 7*(7), 1226–1231.

Forman, C., Goldfarb, A., & Greenstein, S. (2021). Geographic inequality and the internet. *Handbook of digital inequality*. https://doi.org/10.4337/9781788116572.00009

Frischtak, C. R. (2019). Telecommunication and ICT-based services trade. In *Industries without smokestacks, 48*.

Fronteli, M. H., & Pacheco Paladini, E. (2023). Trends, enablers, and barriers for car ownership. *Transportation Research Record, 2677*(1), 1290–1310. https://doi.org/10.1177/03611981221103863

Galich, A., & Stark, K. (2021). How will the introduction of automated vehicles impact private car ownership? *Case studies on transport policy, 9*(2), 578–589. https://doi.org/10.1016/j.cstp.2021.02.012

Grubesic, T. H. (2008). The spatial distribution of broadband providers in the United States: 1999–2004. *Telecommunications Policy, 32*(3–4), 212–233. https://doi.org/10.1016/j.telpol.2008.01.001

Grubesic, T. H., & Murray, A. T. (2002). Constructing the divide: Spatial disparities in broadband access. *Papers in Regional Science, 81*(2), 197–221. https://doi.org/10.1007/s101100100096

Grubesic, T. H., & Murray, A. T. (2004). Waiting for broadband: Local competition and the spatial distribution of advanced telecommunication services in the United States. *Growth and Change, 35*(2), 139–165. https://doi.org/10.1111/j.0017-4815.2004.00243.x

Haining, R., Wise, S., & Ma, J. (1998). Exploratory spatial data analysis in a geographic information system environment. *Journal of the Royal Statistical Society - Series D: The Statistician, 47*(3), 457–469. https://doi.org/10.1111/1467-9884.00147

Hong, J., McArthur, D. P., & Livingston, M. (2019). Can accessing the internet while travelling encourage commuters to use public transport regardless of their attitude? *Sustainability, 11*(12), 3281. https://doi.org/10.3390/su11123281

Hossen, M. A., Talukder, M. R. A., Al Mamun, M. R., Rahaman, H., Paul, S., Rahman, M. M., … Islam, M. N. (2020). Mechanization status, promotional activities and government strategies of Thailand and Vietnam in comparison to Bangladesh. *AgriEngineering, 2*(4), 489–510. https://doi.org/10.3390/agriengineering2040033

ITU. (2022). *Measuring digital development: Facts and figures 2022*. Geneva: ITU Publications.

ITU. (2023). *Facts and figures: Focus on least developed countries*. Geneva: ITU Publications.

Jantavongso, S. (2022). Toward global digital literate citizens: A case of Thailand's aging generation. *The Electronic Journal on Information Systems in Developing Countries, 88*(2), Article e12207. https://doi.org/10.1002/isd2.12207

Jongwanich, J. (2023). Readiness of Thailand towards the digital economy. *Journal of Southeast Asian Economies, 40*(1), 64–95. https://www.jstor.org/stable/27211225.

Jongwanich, J. (2024). *The digital economy in Thailand: Potential and policies* (p. 80).

Karna, D., & Ko, I. (2015). We-intention, moral trust, and self-motivation on accelerating knowledge sharing in social collaboration. In *2015 48th Hawaii international conference on system sciences* (pp. 264–273). IEEE. https://doi.org/10.1109/HICSS.2015.40.

Khan, S., Hwang, G. J., Azeem Abbas, M., & Rehman, A. (2019). Mitigating the urban–rural educational gap in developing countries through mobile technology-supported learning. *British Journal of Educational Technology, 50*(2), 735–749. https://doi.org/10.1111/bjet.12692

Kirkpatrick, R., & Barnawi, O. Z. (2017). Introduction: English language education policy in MENA. *English language education policy in the Middle East and North Africa*, 1–8. https://doi.org/10.1007/978-3-319-46778-8_1

Lee, H. (2022). What drives the performance of Chinese urban and rural secondary schools: A machine learning approach using pisa 2018. *Cities, 123*, Article 103609. https://doi.org/10.1016/j.cities.2022.103609

Lin, F. J. (2008). Solving multicollinearity in the process of fitting regression model using the nested estimate procedure. *Quality and Quantity, 42*(3), 417–426. https://doi.org/10.1007/s11135-006-9055-1

Lin, C. H. A., Lin, H. S., & Hsu, C. P. (2017). Digital divide and income inequality: A spatial analysis. *Review of Economics and Finance, 8*, 31–43.

Ludwig, T., Tolmie, P., & Pipek, V. (2019). From the internet of Things to an internet of practices. *Social Internet of Things*, 33–47. https://doi.org/10.1007/978-3-319-94659-7_3

Mack, E. A., & Grubesic, T. H. (2009). Forecasting broadband provision. *Information Economics and Policy, 21*(4), 297–311. https://doi.org/10.1016/j.infoecopol.2009.08.001

Mack, E. A., Helderop, E., Keene, T., Loveridge, S., Mann, J., Grubesic, T. H., … Gollnow, M. (2022). A longitudinal analysis of broadband provision in tribal areas. *Telecommunications Policy, 102333*. https://doi.org/10.1016/j.telpol.2022.102333

Madianou, M. (2015). Digital inequality and second-order disasters: Social media in the Typhoon Haiyan recovery. *Social Media+ Society, 1*(2). https://doi.org/10.1177/2056305115603386

Marshall, A., Wilson, C. A., & Dale, A. (2023). New pathways to crisis resilience: Solutions for improved digital connectivity and capability in rural Australia. *Media International Australia, 189*(1), 24–42. https://doi.org/10.1177/1329878X231183292

Masonta, M. T. (2023). Towards bridging the digital divide using innovative and community-based interventions. In *2023 IST-africa conference (IST-Africa)* (pp. 1–8). IEEE. https://doi.org/10.23919/IST-Africa60249.2023.10187842.

Mehta, N. (2017). Relationship between farm and allied non-farm sectors: Issues relating to linkages and future of agriculture. *Indian Journal of Agricultural Economics, 72*(3), 484–495.

Murphy, A., & Arciuli, J. (2024). Digital reading comprehension instruction in English for children with English as an additional language: A systematic review. *Journal of Research in Reading*. https://doi.org/10.1111/1467-9817.124483

Myovella, G., Karacuka, M., & Haucap, J. (2021). Determinants of digitalization and digital divide in sub-saharan african economies: A spatial durbin analysis. *Telecommunications Policy, 45*(10), Article 102224. https://doi.org/10.1016/j.telpol.2021.102224

Nanthakorn, B., Ractham, P., & Suwannakart, P. (2022). Digital inequalities at high schools in Thailand: A survey-based exploration leading to expert-backed bridging strategies. In *Proceedings of the 22nd international conference on electronic business (ICEB)* (pp. 76–84).

National Economic and Social Development Board. (2018). *Report on poverty and inequality in income -om regional perspectives in Thailand*. Bangkok, National Economic and Social Development Board.

National Science and Technology Development Agency. (2011). Guidelines for registering multipurpose agricultural vehicles in Thailand. Pathum Thani: Nstda. Retrieved from https://www.nstda.or.th/asean-india-dl/sites/default/files/20110823-policy-registration-of-the-vehicle.pdf.

National Statistical Office of Thailand. (2018). *Poverty rate in Thailand by region over 10 years (Million people)*. Bangkok (National Statistical Office of Thailand.

National Statistical Office of Thailand. (2021). The 2021 (quarter 3) household survey on the use of information and communication technology. *Bangkok, National Statistical Office of Thailand*.

Nguyen, T. T., Nguyen, T. T., & Grote, U. (2022). Internet use, natural resource extraction and poverty reduction in rural Thailand. *Ecological Economics, 196*, Article 107417. https://doi.org/10.1016/j.ecolecon.2022.107417

Ngwenya, S. O., Heymann, R., Swart, T. G., & Lysko, A. (2023). A comparative analysis of urban and rural broadband penetration and access trends in South Africa. In *Proceedings of the international conferences on ICT, society, and human beings 2023, ICT 2023; e-health 2023, EH 2023; connected smart cities 2023, CSC 2023; and big data analytics, data mining and computational intelligence 2023, BigDaCI 2023* (pp. 64–72). Porto: IADIS Press. https://hdl.handle.net/10210/504783.

Niehaves, B., & Plattfaut, R. (2014). Internet adoption by the elderly: Employing IS technology acceptance theories for understanding the age-related digital divide. *European Journal of Information Systems, 23*(6), 708–726. https://doi.org/10.1057/ejis.2013.19

Nielsen, M. M., Rohman, I. K., & Lopes, N. V. (2018). Empirical analysis of the current digital divides since 2010. In *Proceedings of the 11th international conference on theory and practice of electronic governance* (pp. 616–625). https://doi.org/10.1145/3209415.3209493

Nishida, T., Pick, J. B., & Sarkar, A. (2014). Japan s prefectural digital divide: A multivariate and spatial analysis. *Telecommunications Policy, 38*(11), 992–1010. https://doi.org/10.1016/j.telpol.2014.05.004

Odongo, A. O. (2014). Research on ICT development in Africa: Countries focusing on benefits, trends, challenges and solutions. In *Proceedings of the 8th international conference on theory and practice of electronic governance* (pp. 379–382). https://doi.org/10.1145/2691195.2691226

OECD. (2001). Understanding the digital divide, OECD digital economy papers No. 49. https://doi.org/10.1787/236405667766.

OECD. (2022). In *Open and connected government review of Thailand, OECD public governance reviews*. Paris: OECD Publishing. https://doi.org/10.1787/e1593a0c-en.

Office of the National Economic and Social Development Council of Thailand. (2021). *Thailand's poverty and inequality Report 2020*. Bangkok. https://www.nesdc.go.th/ewt_dl_link.php?nid=11972.

Ohemeng, F. L. K., & Ofosu-Adarkwa, K. (2014). Overcoming the digital divide in developing countries: An examination of Ghana's strategies to promote universal access to information communication technologies (ICTs). *Journal of Developing Societies, 30*(3), 297–322. https://doi.org/10.1177/0169796X14536970

Ordóñez, K., García, P., & Ramón, M. (2016). Internet impact on economic development of member countries of the Andina Community. In *2016 11th iberian conference on information systems and technologies (CISTI)* (pp. 1–5). IEEE. https://doi.org/10.1109/CISTI.2016.7521583.

Oyedemi, T., & Mogano, S. (2018). The digitally disadvantaged: Access to digital communication technologies among first year students at a rural South African University. *Africa Education Review, 15*(1), 175–191. https://doi.org/10.1080/18146627.2016.1264866

Pederzoli, D. (2019). How communities affect the technology acceptance model in the retail sector. *The impact of information and communication technologies on organizations and society*. ICT for a Better Life and a Better World. https://doi.org/10.1007/978-3-030-10737-6_26

Peraphan, J., Hermann, K., & Markus, M. (2017). The conundrum of the motorcycle in the mix of sustainable urban transport. *Transportation Research Procedia, 25*, 4869–4890. https://doi.org/10.1016/j.trpro.2017.05.365

Phoompanich, S., Barr, S., & Gaulton, R. (2019). Development of geospatial techniques for natural hazard risk assessment in Thailand. The International Archives of the Photogrammetry. *Remote Sensing and Spatial Information Sciences, 42*, 315–322. https://doi.org/10.5194/isprs-archives-XLII-3-W8-315-2019

Pick, J. B., & Azari, R. (2011). A global model of utilization of technology based on governmental, social, economic, and business investment factors. *Journal of Management Information Systems, 28*(1), 51–85. https://doi.org/10.2753/MIS0742-1222280103

Pick, J. B., & Nishida, T. (2015). Digital divides in the world and its regions: A spatial and multivariate analysis of technological utilization. *Technological Forecasting and Social Change, 91*, 1–17. https://doi.org/10.1016/j.techfore.2013.12.026

Pick, J. B., & Sarkar, A. (2015). *The global digital divides: Explaining change*. Springer.

Pick, J. B., & Sarkar, A. (2016). Theories of the digital divide: Critical comparison. *2016 the 49th Hawaii international conference on system sciences*. https://doi.org/10.1109/HICSS.2016.484

Pick, J. B., Sarkar, A., & Johnson, J. (2015). United States digital divide: State level analysis of spatial clustering and multivariate determinants of ICT utilization. *Socio-Economic Planning Sciences, 49*, 16–32. https://doi.org/10.1016/j.seps.2014.09.001

Pick, J. B., Sarkar, A., & Parrish, E. (2021). The Latin American and caribbean digital divide: A geospatial and multivariate analysis. *Information Technology for Development, 27*(2), 235–262. https://doi.org/10.1080/02681102.2020.1805398

Porter, C. E., & Donthu, N. (2006). Using the technology acceptance model to explain how attitudes determine Internet usage: The role of perceived access barriers and demographics. *Journal of Business Research, 59*(9), 999–1007. https://doi.org/10.1016/j.jbusres.2006.06.003

Punyasavatsut, A. (2019). The effect of internet penetration on economic growth. *Kasetsart Journal of Social Sciences, 40*(1), 256–261.

Qadikolaei, M. R., Zali, N., & Soltani, A. (2024). Spatiotemporal investigation of the digital divide, the case study of Iranian Provinces. *Environment, Development and Sustainability, 26*(1), 869–884. https://doi.org/10.1007/s10668-022-02738-0

Rice, R. E., & Pearce, K. E. (2015). Divide and diffuse: Comparing digital divide and diffusion of innovations perspectives on mobile phone adoption. *Mobile Media & Communication, 3*(3), 401–424. https://doi.org/10.1177/2050157915590469

Rogers, E. M. (1983). *Diffusion of innovations* (3rd ed.). The Free Press.

Sarkar, A., Pick, J. B., & Johnson, J. (2015). Africa's digital divide: Geography, policy, and implications. In *2015 regional conference of the international telecommunications society (ITS): "The intelligent world: Realizing hopes, overcoming challenges", Los Angeles, USA, 25th-28th october 2015, international telecommunications society (ITS), calgary*. https://www.econstor.eu/handle/10419/146339.

Setthasuravich, P., & Kato, H. (2020). The mediating role of the digital divide in outcomes of short-term transportation policy in Thailand. *Transport Policy, 97*, 161–171. https://doi.org/10.1016/j.tranpol.2020.07.008

Setthasuravich, P., & Kato, H. (2022). Does the digital divide matter for short-term transportation policy outcomes? A spatial econometric analysis of Thailand. *Telematics and Informatics, 72*, Article 101858. https://doi.org/10.1016/j.tele.2022.101858

Sipior, J. C., Ward, B. T., & Connolly, R. (2011). The digital divide and t-government in the United States: Using the technology acceptance model to understand usage. *European Journal of Information Systems, 20*(3), 308–328. https://doi.org/10.1057/ejis.2010.64

Song, Z., Wang, C., & Bergmann, L. (2020). China's prefectural digital divide: Spatial analysis and multivariate determinants of ICT diffusion. *International Journal of Information Management, 52*, Article 102072. https://doi.org/10.1016/j.ijinfomgt.2020.102072

Sujarwoto, S., & Tampubolon, G. (2016). Spatial inequality and the Internet divide in Indonesia 2010–2012. *Telecommunications Policy, 40*(7), 602–616. https://doi.org/10.1016/j.telpol.2015.08.008

Sun, S., Guo, L., Yang, S., & Cao, J. (2024). Exploring the contributions of Ebike ownership, transit access, and the built environment to car ownership in a developing city. *Journal of Transport Geography, 116*, Article 103834. https://doi.org/10.1016/j.jtrangeo.2024.103834

Szeles, M. R. (2018). New insights from a multilevel approach to the regional digital divide in the European Union. *Telecommunications Policy, 42*(6), 452–463. https://doi.org/10.1016/j.telpol.2018.03.007

Thakur, K. K., & Prasad, R. (2021). Wi-Fi for affordable broadband & 5G in rural areas. *Journal of Mobile Multimedia*, 225–244. https://doi.org/10.13052/jmm1550-4646.171312

van Deursen, A. J. (2020). Digital inequality during a pandemic: Quantitative study of differences in COVID-19–related internet uses and outcomes among the general population. *Journal of Medical Internet Research, 22*(8), Article e20073. https://doi.org/10.2196/20073

van Deursen, A. J., van der Zeeuw, A., de Boer, P., Jansen, G., & van Rompay, T. (2021). Digital inequalities in the internet of Things: Differences in attitudes, material access, skills, and usage. *Information, Communication & Society, 24*(2), 258–276. https://doi.org/10.1080/1369118X.2019.1646777

van Dijk, J. (2020). *The digital divide*. Cambridge: Polity press.

van Laar, E., van Deursen, A. J., van Dijk, J. A., & de Haan, J. (2019). Determinants of 21st-century digital skills: A large-scale survey among working professionals. *Computers in Human Behavior, 100*, 93–104. https://doi.org/10.1016/j.chb.2019.06.017

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly, 27*(3), 425–478. https://doi.org/10.2307/30036540

Vincent, R. C. (2016). The internet and sustainable development: Communication dissemination and the digital divide. *Perspectives on Global Development & Technology, 15*(6), 605–637. https://doi.org/10.1163/15691497-12341410

Visessri, S., & Ekkawatpanit, C. (2020). Flood management in the context of climate and land-use changes and adaptation within the Chao Phraya River basin. *Journal of Disaster Research, 15*(5), 579–587. https://doi.org/10.20965/jdr.2020.p0579

Vural, C. A., Roso, V., Halldórsson, Á., Ståhle, G., & Yaruta, M. (2020). Can digitalization mitigate barriers to intermodal transport? An exploratory study. *Research in Transportation Business & Management, 37*, Article 100525. https://doi.org/10.1016/j.rtbm.2020.100525

Wang, T., Guo, X., & Wu, T. (2021). Social capital and digital divide: Implications for mobile health policy in developing countries. *Journal of Healthcare Engineering, 2021*. https://doi.org/10.1155/2021/6651786

Wang, Q., Xia, X., Lan, S., & Li, M. (2023). Rural digital infrastructure and labor market: Evidence from universal telecommunication service. *Asian Economic Journal, 37*(3), 293–325. https://doi.org/10.1111/asej.12306

Wilson, J. (2021). Thailand's Rao Chana program as told by those who participated [online] https://www.thaienquirer.com/24415/thailands-rao-chana-program-critiqued-by-participants-of-the-subsidy-scheme/. (Accessed 14 December 2021).

Wims, P. (2014). The potential of agribusiness extranets to improve communications with farmer clients: A case study from Ireland. *International Journal of Electronic Business, 11*(1), 24–41. https://doi.org/10.1504/IJEB.2014.057899

Wongyai, K., & Huttasin, N. (2022). Self-Adaptation strategy of internet cafe to build up competitive advantage in mueang district, ubon ratchathani province. *Journal of Management Science, Ubon Ratchathani University, 11*(2), 1–17. Retrieved from https://so03.tci-thaijo.org/index.php/jms_ubu/article/view/252522.

Yuan, Y. (2021). Support social connectedness through technology-mediated shared activities. In *Companion publication of the 2021 conference on computer supported cooperative work and social computing* (pp. 307–309). https://doi.org/10.1145/3462204.3481801

Zhao, P., & Bai, Y. (2019). The gap between and determinants of growth in car ownership in urban and rural areas of China: A longitudinal data case study. *Journal of Transport Geography, 79*, Article 102487. https://doi.org/10.1016/j.jtrangeo.2019.102487

Zheng, Y. Y., Zhu, T. H., & Wei, J. I. A. (2022). Does internet use promote the adoption of agricultural technology? Evidence from 1 449 farm households in 14 Chinese provinces. *Journal of Integrative Agriculture, 21*(1), 282–292. https://doi.org/10.1016/S2095-3119(21)63750-4

Zhou, C., Zhang, D., & Chen, Y. (2022). Theoretical framework and research prospect of the impact of China's digital economic development on population. *Frontiers in Earth Science, 10*, Article 988608. https://doi.org/10.3389/feart.2022.988608